MODERN PHYSICS

DUAL NATURE OF MATTER AND RADIATION

dual nature k teen part h PEE, D B aur radiation force blah blah blah...

Photoelectric Effect

E(eV) = 12400 / λ(Å) or 1240 / λ(nm) {1240//1242} :)

Introduction to Photoelectric Effect, Photoelectrons, and Photo Current

- Definition of Photoelectric Effect:

When electromagnetic radiation (light) of a frequency greater than or equal to a minimum characteristic value falls on a metallic surface, electrons are liberated from the metal surface. This phenomenon is known as the photoelectric effect.

- Photoelectrons:

The electrons emitted from the metal surface during the photoelectric effect are called photoelectrons.

- Photo Current:

The rate of flow of emitted photoelectrons across the circuit constitutes an electric current termed the photo current.

Threshold Frequency (ν_0), Work Function (ϕ), and Cut-off Wavelength (λ_0)

Threshold Frequency (ν_0):

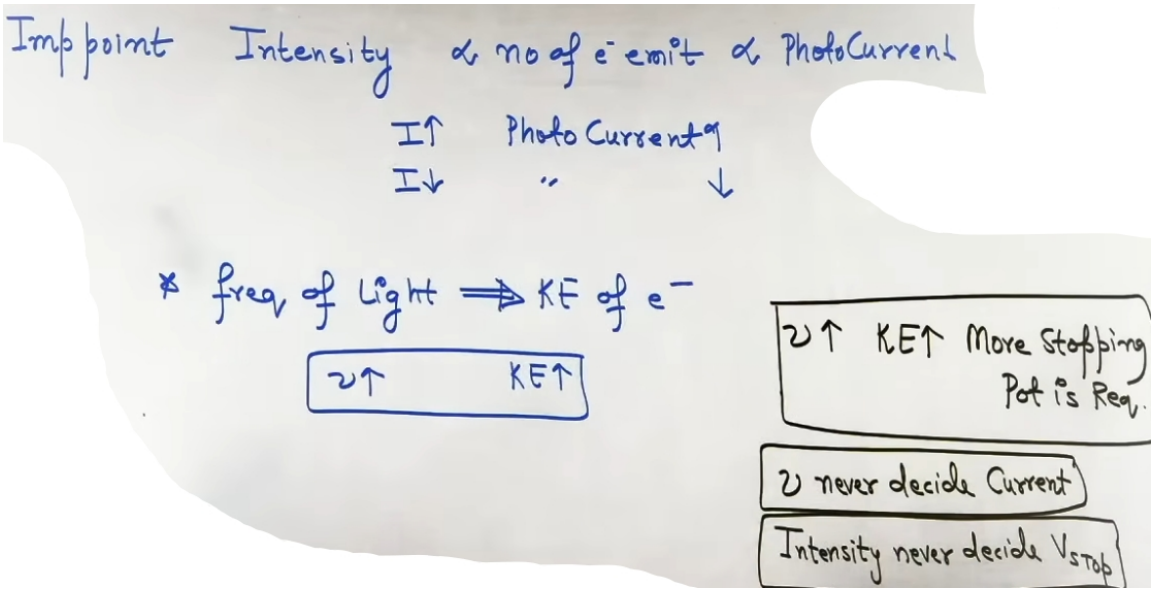
- The minimum frequency of incident light required to initiate photoelectron emission from a given metal surface.
- If the incident frequency $\nu < \nu_0$, no photoelectric emission occurs regardless of light intensity or exposure duration.
- Threshold frequency is strictly metal-dependent (varies from metal to metal).

Work Function (ϕ):

- The minimum energy required to remove a bound electron from a metal surface.
- Formula: $\phi = h * \nu_0 = (h * c) / \lambda_0$. # (1 eV = 1.6×10^{-19} J)
- SI Unit: Joule (J), commonly expressed in electron-volts (eV).
- Work function depends exclusively on the nature of the metal surface.

Cut-off Wavelength / Threshold Wavelength (λ_0):

- The maximum wavelength of incident light up to which the photoelectric effect will occur.
- If incident wavelength $\lambda > \lambda_0$, photoelectric emission ceases completely.
  Formula: $\lambda_0 = c / \nu_0 = (h * c) / \phi$.



Experimental Observations and Graphs

1. Effect of Light Intensity on Photo Current:

- When frequency $\nu > \nu_0$ is kept constant, the photo current is directly proportional to the intensity of incident light.
- Increasing light intensity increases the number of photoelectrons emitted per second, thereby increasing the photo current linearly.

2. Instantaneous Emission Process:

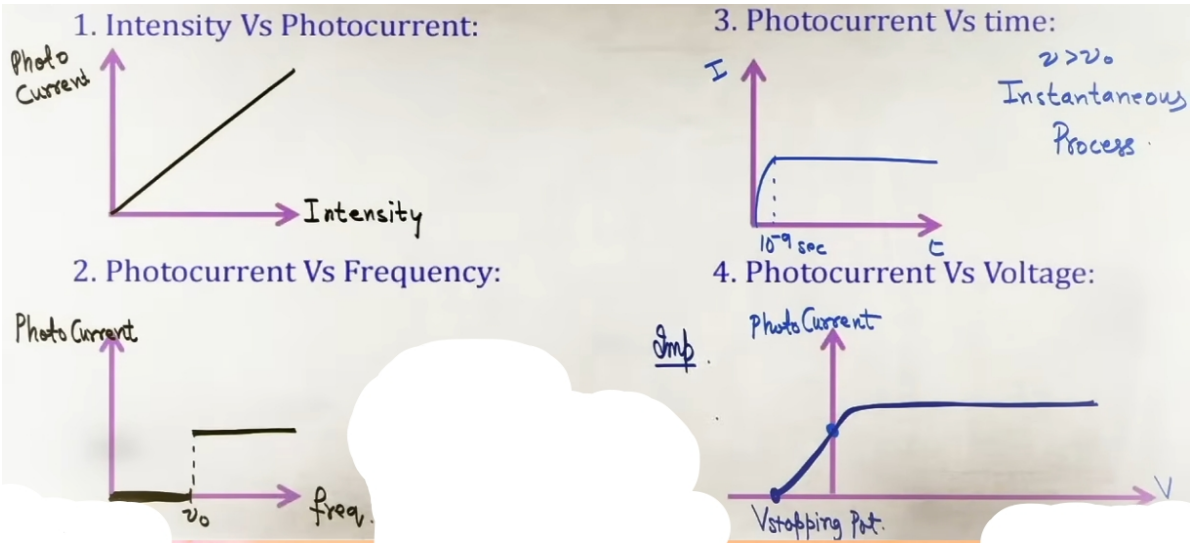
- The emission of photoelectrons is an instantaneous process.
- The time interval between the incidence of radiation and the emission of photoelectrons is less than 10^{-9} seconds.

3. Effect of Light Frequency on Photo Current:

- Photo current does not depend on incident light frequency once the frequency exceeds threshold frequency ν_0.
- If $\nu < \nu_0$, photo current is strictly zero.
- Once $\nu \geq \nu_0$, photo current rises instantaneously to a constant saturation value dictated by light intensity.

4. Effect of Applied Collector Potential (Voltage) on Photo Current:

- At zero applied voltage (V = 0), a non-zero photo current exists because emitted electrons possess initial kinetic energy and scatter toward the collector.
- As positive potential (accelerating voltage) on the collector increases, scattered photoelectrons are pulled in, causing a slight initial increase in photo current.
- Beyond a certain positive potential, photo current becomes constant and stops increasing despite further increases in positive voltage.



Saturation Current and Stopping Potential (Cut-off Voltage)

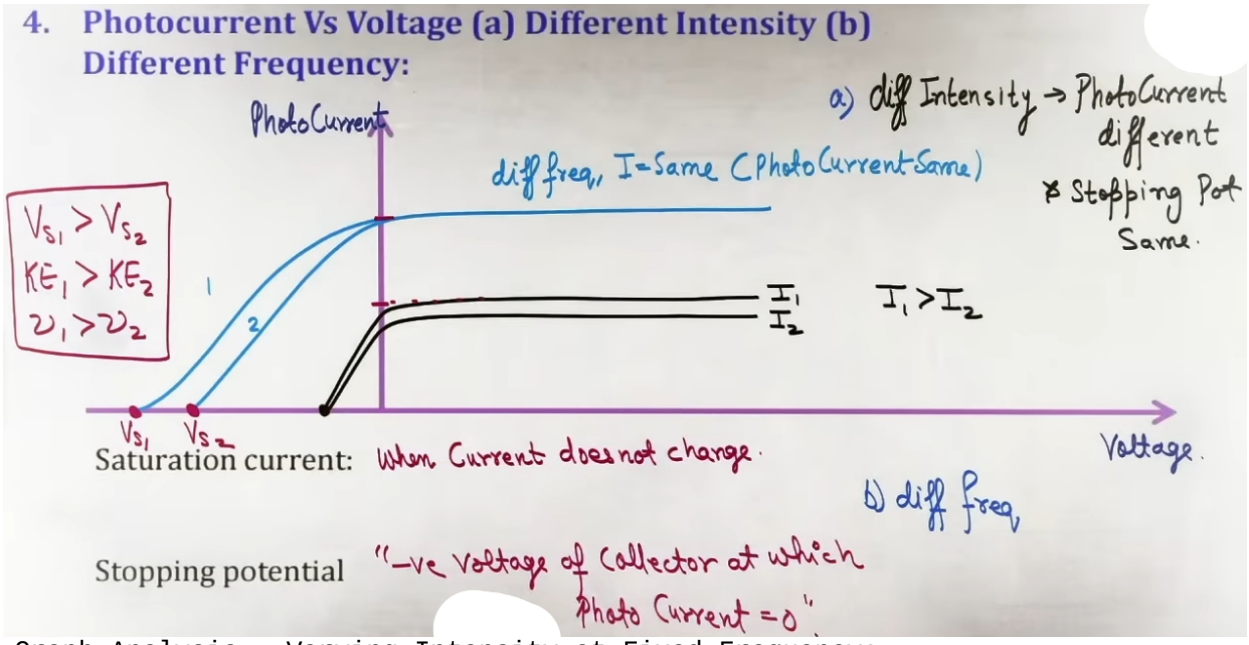
Saturation Current:

- The maximum constant photo current achieved when all photoelectrons emitted per second from the emitter plate reach the collector plate.
- Saturation current is decided exclusively by light intensity (higher intensity gives higher saturation current).

Stopping Potential (V_s / Cut-off Voltage):

- When a negative (retarding) potential is applied to the collector plate relative to the emitter, emitted photoelectrons experience electrostatic repulsion.
- Stopping Potential (V_s) is the minimum negative potential applied to the collector plate at which the photo current is reduced to exactly zero.
- At stopping potential V_s, even the most energetic photoelectron possessing maximum kinetic energy (K_max) is stopped right before reaching the collector.

"Fundamental Relation: K_max = e * V_s."



- Graph Analysis - Varying Intensity at Fixed Frequency:
  - Saturation currents differ (higher intensity $I_1 > I_2$ produces higher saturation current).
  - Stopping potential V_s remains EXACTLY THE SAME for all intensities because frequency is constant.

- Graph Analysis - Varying Frequency at Fixed Intensity:
  - Saturation current remains EXACTLY THE SAME because intensity is constant.
  - Stopping potentials differ ($|V_{s1}| > |V_{s2}|$ for $\nu_1 > \nu_2$), meaning higher frequency requires a more negative stopping potential.

Failure of Classical Wave Theory of Light

Classical wave theory failed to explain the photoelectric effect due to three irreconcilable contradictions with experimental observations:

1. The Time Lag Problem:

  Wave theory assumes light energy is spread continuously across a wavefront. Calculations implied an electron on a metal surface would take a considerable time gathering wave energy continuously until accumulating enough energy equal to the work function.

  Contradiction: Experimentally, emission is instantaneous (time lag < 10^{-9} s).

2. The Intensity vs Kinetic Energy Problem:

  Wave theory asserts light intensity represents wave amplitude and energy per unit area per unit time.

  Increasing intensity should deliver more wave energy per electron, imparting higher kinetic energy and increasing stopping potential.

  Contradiction: Experimentally, kinetic energy K_max and stopping potential V_s are independent of intensity and depend solely on frequency.

3. The Threshold Frequency Problem:

  Wave theory predicts light of any frequency, if made sufficiently intense or shone for long enough, should accumulate enough energy to eject electrons.

  Contradiction: Experimentally, no photoelectrons are emitted if $\nu < \nu_0$, regardless of intensity or exposure time.

INTUITION AND CONCEPTUAL UNDERSTANDING

- Mental Model - Quanta / Photons vs Continuous Waves:

  Classical wave theory treats light energy like a continuous hose stream spraying water over an area. An electron would have to wait for enough water to fill its bucket before jumping out.

  Einstein's quantum model treats light like hail or bullets (photons). Energy arrives in localized packets E = h * ν. An electron gets hit by a single bullet instantaneously.

- *One-to-One Collision Principle:*
  Photoelectric emission is a 1-to-1 interaction between one incident photon and one bound electron inside the metal.
  If the single incoming photon possesses energy $h * \nu \geq \phi$, the electron absorbs it instantaneously and escapes.
  If $h * \nu < \phi$, the photon cannot eject the electron, no matter how many other sub-threshold photons strike the surface simultaneously.

- Real-Life Analogy for Intensity vs Frequency:
  - Intensity = Number of photons delivered per second (brightness).
  - Frequency = Energy contained inside each individual photon.
  - Analogy: Dropping 100 ping-pong balls (high intensity, low frequency) onto a glass pane will not break it. Dropping a single steel cannonball (low intensity, high frequency) shatters the glass instantly.

FORMULA SHEET

- Work Function:
  $\phi = h * \nu_0 = (h * c) / \lambda_0$
  Units: Joule (J) or Electron-Volt (eV)
  Condition: Surface property of a specific metal.

- Threshold Frequency and Cut-off Wavelength:
  $\nu_0 = \phi / h$
  $\lambda_0 = (h * c) / \phi = c / \nu_0$

- Einstein's Photoelectric Equation:
  $E_{incident} = \text{Work Function} + K_{max}$
  $h*\nu = \phi + K_{max}$
  $h\nu = h\nu_0 + (1/2)m(v_{max})^2$
Condition: K_{max} represents the MAXIMUM possible kinetic energy of emitted photoelectrons.

- Stopping Potential Equations:
  $K_{max} = e*V_s$
  $eV_s = h*\nu - \phi$
  $V_s = (h/e)*\nu - (\phi/e)$

Parameter Dependence Table:

Parameter Changed	Impact on Photo Current / Saturation Current	Impact on K_{max}	Impact on Stopping Potential V_s
Increase Intensity (Fixed ν)	Increases Linearly	No Change	No Change
Decrease Intensity (Fixed ν)	Decreases Linearly	No Change	No Change
Increase Frequency (Fixed Intensity)	No Change	Increases Linearly	Becomes More Negative
Decrease Frequency ($\nu > \nu_0$)	No Change	Decreases Linearly	Becomes Less Negative
Frequency $\nu < \nu_0$	Drops to Zero	Zero (No Emission)	Zero

PROBLEM-SOLVING FRAMEWORK

On mixed Light Frequencies Incident on Metal, light equation contains multiple frequencies (e.g. $E = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t$)
1. Identify highest frequency $\nu_{max} = \omega_{max} / 2\pi$.
2. Stopping potential is determined EXCLUSIVELY by ν_{max}. Highest frequency component determines K_{max} and V_s.

$E(eV) = 12400 / \lambda(Å)$ or $1240 / \lambda(nm)$ # ($1\ eV = 1.6\times10^{-19}\ J$)

JEE ADVANCED INSIGHTS

1. Universal Slope of V_s vs ν Graphs:
The slope of the Stopping Potential (V_s) versus Frequency (ν) graph is $m = h / e$. Since Planck's constant h and elementary charge e are universal constants, the slope of V_s vs ν is identical for every photosensitive metal.

2. Shifted Origin Traps in Graphs:
Exams frequently shift the origin of the V_s vs ν graph (e.g. starting the X-axis at $\nu = 4 \times 10^{14}$ Hz instead of 0). Do not blindly read the visual axis intersection as ν_0. Always check the numerical coordinate values at the intercept.

3. Maximum vs Range of Kinetic Energies:
Einstein's equation $h * \nu = \phi + K_{max}$ yields the MAXIMUM kinetic energy of photoelectrons (emitted from the top surface layer without collisions). Internal electrons undergo multiple scattering collisions inside the bulk metal before escaping, losing varying amounts of energy. Thus, emitted photoelectrons possess a continuous range of kinetic energies: $0 \leq K \leq K_{max}$.

HIGH-YIELD REVISION SHEET
MUST REMEMBER:

Key Formulas:
- Work Function: $\phi = h\nu_0 = (hc)/\lambda_0$
- Einstein's Equation: $h\nu = \phi + K_{max} = h\nu_0 + e(V_s)$
- Stopping Potential: $V_s = (h/e)*\nu - (\phi/e)$
- Maximum Kinetic Energy: $K_{max} = e*V_s = (1/2)m(v_{max})^2$

Key Observations:
1. Intensity ∝ Number of photoelectrons emitted per second ∝ Saturation photo current.
2. Frequency ∝ Photon Energy ∝ Maximum Kinetic Energy (K_{max}) ∝ Stopping Potential Magnitude ($|V_s|$).
3. Emission is instantaneous ($< 10^{-9}$ s time lag).
4. Emitted photoelectrons have kinetic energies ranging continuously from 0 to K_{max}.
5. Slope of V_s vs ν graph is universally constant = h / e.

EINSTEIN'S QUANTUM/PHOTON THEORY AND PHOTOELECTRIC EQUATION
Photon Hypothesis:
- Light/radiation does not transfer energy continuously like classical waves, but propagates in localized, discrete energy packets called quanta or photons.
- Energy of a single photon: $E = h * \nu = hc/\lambda$
  where h is Planck's constant (6.63×10^{-34} J·s), ν is frequency, c is speed of light (3×10^8 m/s), and λ is wavelength.
- Total energy for N photons: $E_{total} = N*h*\nu = N*(hc)/\lambda$.

1-to-1 Photon-Electron Interaction Principle:
- The photoelectric interaction is a discrete, "instantaneous" 1-to-1 collision between one incident photon and one bound electron inside the metal.
- A photon gives up its entire energy to a single electron upon collision. It cannot divide its energy among multiple electrons, nor can an electron accumulate energy from multiple sub-threshold photons.

- The intensity of light is directly proportional to no. of photons, hence if intensity increases the no of photon incident increases which leads to more emission of electrons (riso of photocurrent)

Conservation Laws in Photoelectric Emission:
- In every photon-electron collision, total energy and total linear momentum are conserved.

Einstein's Photoelectric Equation:
- By conservation of energy:
  E_incident = Work Function (ϕ) + Maximum Kinetic Energy (K_max)
  $h*\nu = h*\nu_0 + (1/2)*m*(v_{max})^2$
  $(h*c)/\lambda = (h*c)/\lambda_0 + (1/2)*m*(v_{max})^2$

Relation to Stopping Potential (V_s):
- The maximum kinetic energy K_max equals the electrostatic work done by retarding potential V_s to stop the fastest electron:
  K_max = e * V_s
  $(h*c)/\lambda = (h*c)/\lambda_0 + e*V_s$

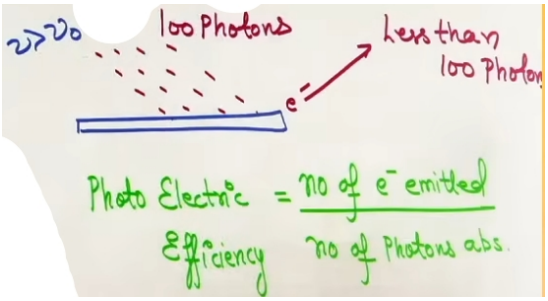
- If the frequency of light increases the kinetic energy of photo electron increases, hence the stopping potential. # explanation on next part

- No. of photons incident is not equal to no of electron emitted.

- For a given incident frequency, the kinetic energy of emitted electron is not same, only few electrons have maximum kinetic energy. # exp. on next part

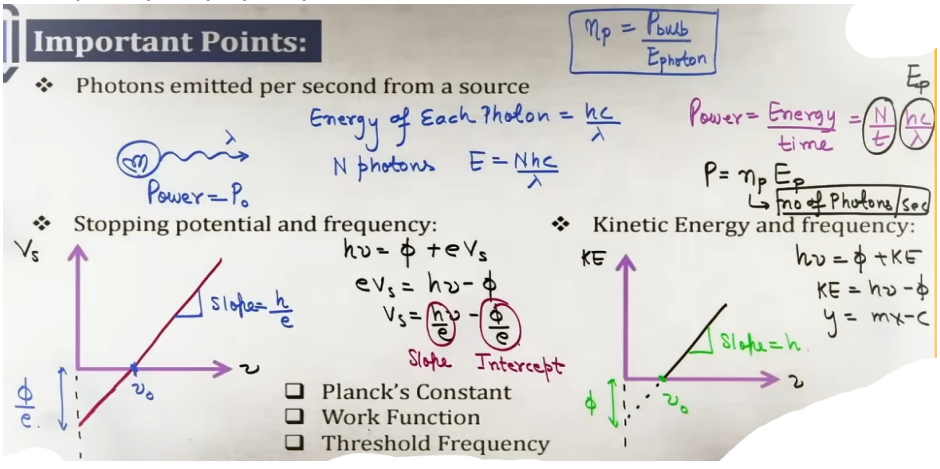
PHOTOELECTRIC EFFICIENCY (QUANTUM YIELD) AND PHOTON EMISSION RATE / POWER FORMULA

Photoelectric Efficiency (Quantum Yield):
- Defined as the ratio of the number of emitted photoelectrons per second to the number of incident photons absorbed per second:
- Photoelectric Efficiency = (Number of photoelectrons emitted per second / Number of photons absorbed per second) * 100%



  In real metal surfaces, photoelectric efficiency is very low (typically less than 1%).
  Reason: Most absorbed photons interact with bulk electrons beneath the surface. These electrons lose kinetic energy through multiple collisions with the metal lattice before escaping, leaving only a small fraction to emerge with maximum kinetic energy K_max.
  Range of kinetic energies of emitted photoelectrons: $0 \le K \le K_{max}$.

Photon Emission Rate (n_p) and Source Power Formula:
- Let a monochromatic light source of power P emit light of wavelength λ and frequency ν.
  Energy of each photon: $E_p = h * \nu = (h * c) / \lambda$
  Energy of N photon: $E = Nh\nu = (N)(hc)/\lambda$
  Power (P) = Energy/time = $(N)h*\nu/t = (N/t)(h*\nu) = n_p*E_p$
  where $n_p = N/t$ is the number of photons emitted per second.
  Photon Emission Rate Formula:
  $n_p = P / E_p = (P*\lambda)/(h*c)$



Q. Given figure shows few data points in a photo electric effect experiment for a certain metal. The minimum energy for ejection of electron from its surface is (Plancks constant h = 6.62×10^{-34} J. S)

[JEE Mains 2020 7 September (Morning)]

Ans : B

A) 2.59eV

B) 2.27eV

C) 1.93eV

D) 2.10eV

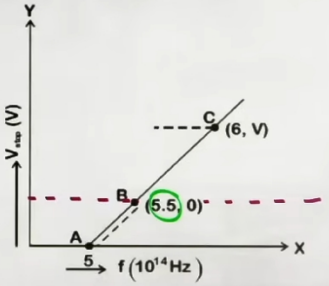
ϕ = work function

$\phi = h\upsilon_0$

$\phi = 6.62\times10^{-34} \times 5.5\times10^{14}$

In eV.

$\phi = \dfrac{6.62\times10^{-34}\times5.5\times10^{14}}{1.6\times10^{-19}}$



Q. The electric field of light wave is given as $E = 10^{-3}\cos\left(\frac{2\pi x}{5\times10^{-7}} - 2\pi \times 6 \times 10^{14}t\right)\hat{x}$ NC^{-1}. This light falls on a metal plate to work function 2eV. The stopping potential of the photoelectrons is [Main 2019]

(Given, E (in eV) = $\frac{12375}{\lambda\,(\text{in Å})}$)

Ans : A

A) 0.48 eV

B) 0.72 eV

C) 2.0 eV

D) 2.48 eV

$\frac{2\pi}{\lambda}$ $\quad 2\pi\upsilon$

$E = E_0\cos(Kx - \omega t)$

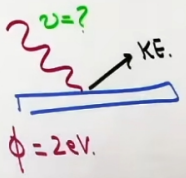
$E = 10^{-3}\cos\left(\frac{2\pi x}{5\times10^{-7}} - 2\pi\times6\times10^{14}t\right)$

λ

$2\pi\times6\times10^{14} = 2\pi\upsilon$

$\upsilon = 6\times10^{14}$ Hz

Incident freq.



$E_i = \phi + eV_s$

$= 2eV + eV_s$

$\lambda_i = 5\times10^{-7} = 5000$ Å

$E_i = \frac{12375}{5000}$ eV

Q. The magnetic field associated with a light wave is given at the origin, by $B = B_0[\sin(3.14\times10^7)ct + \sin(6.28\times10^7)ct]$. If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of the photoelectrons?

(Take, c = 3×10^8 ms^{-1} and h = 6.6×10^{-34} J-s)

[Main 2019]

Ans : A

A) 7.72 eV

B) 6.82 eV

C) 8.52 eV

D) 12.5 eV

ω_2 $(2\pi\times10^7 s)$

$B = B_0[\sin(3.14\times10^7)ct + \sin(6.28\times10^7)ct]$

ω_1 υ_1 $\quad$ ω_2 υ_2 (more)

$\omega = 2\pi\upsilon$

$\phi = 4.7$ eV.

$\upsilon \rightarrow$ KE.

$E_i = \phi + KE$

$E_i = 4.7eV + KE$

$KE_e \rightarrow$ more.

$\omega_2 = (2\pi\times10^7) = 2\pi\upsilon_2$

Since it is more than ω_1 its KE_e will be More.

$\upsilon_2 = 10^7$

$\upsilon_2 = 3\times10^8\times10^7$

$E_i = h\upsilon = \frac{6.6\times10^{-34}\times3\times10^{15}}{1.6\times10^{-19}}$

$E_i =$ (eV)

$E_i = \phi + KE$

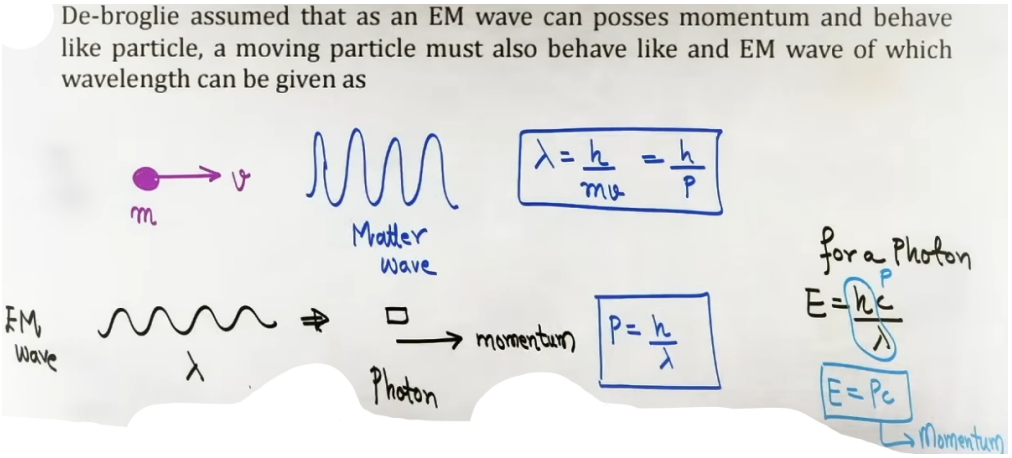
DE BROGLIE MATTER WAVE HYPOTHESIS AND WAVELENGTH CALCULATIONS

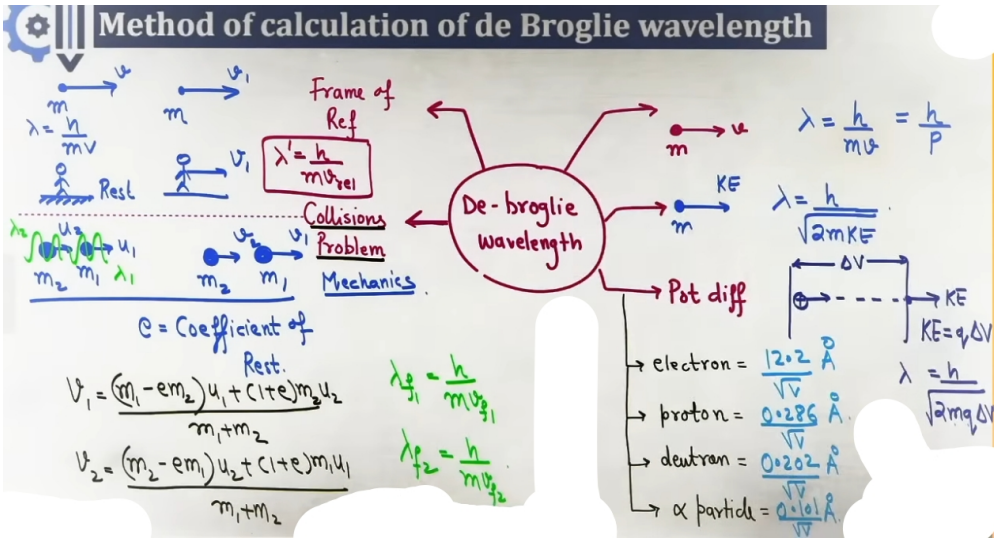
De Broglie Hypothesis:

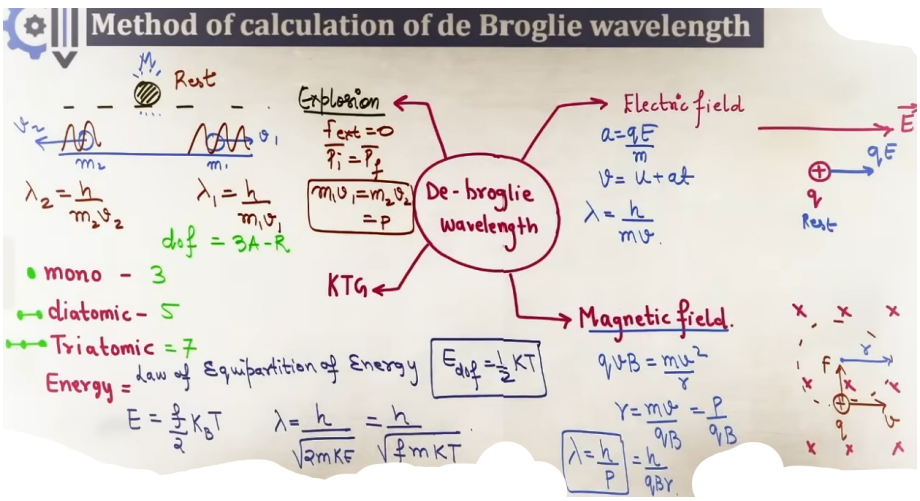
- Nature exhibits fundamental symmetry. If light waves exhibit particle-like properties (photons), moving material particles must exhibit wave-like properties (matter waves).
  Associated De Broglie Wavelength (λ):
  λ = h/p = h/(mv)
  where p is particle momentum, m is mass, and v is velocity.

Photon Momentum:

- For a photon of energy E = hc/λ, rest mass is zero, but momentum is:
  p = E/c = h/λ.







Q. A particle P is formed due to a completely inelastic collision of particles x and y having de-Broglie wavelengths λ_x and λ_y, respectively. If x and y were moving in opposite directions, then the de-Broglie wavelength of P is

(Main 2019)

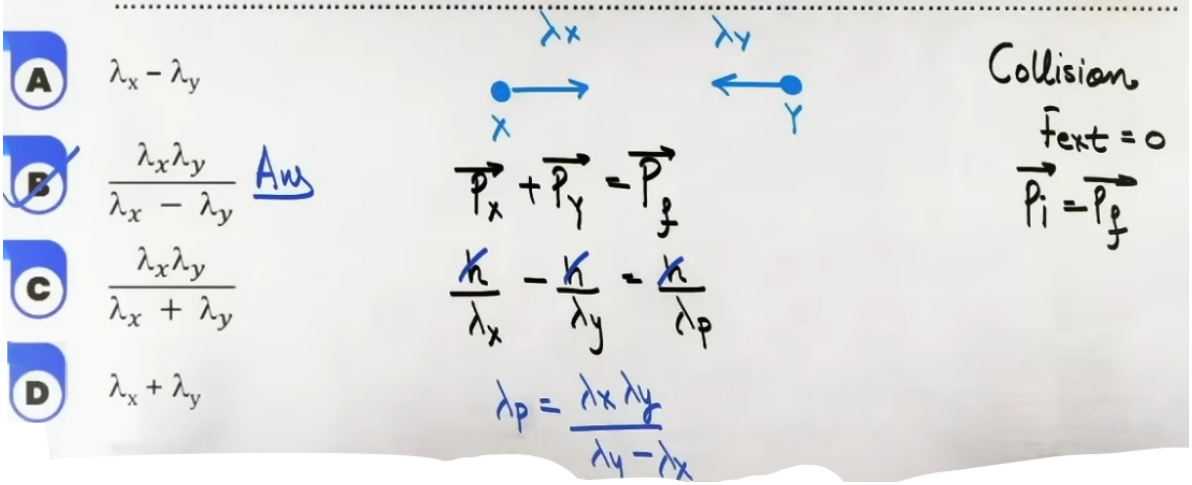
Ans : B

A. $\lambda_x - \lambda_y$

B. $\frac{\lambda_x \lambda_y}{\lambda_x - \lambda_y}$ Ans

C. $\frac{\lambda_x \lambda_y}{\lambda_x + \lambda_y}$

D. $\lambda_x + \lambda_y$



Q. An electron of mass m with an initial velocity $\vec{V} = V_0\hat{i}$ $(V_0 > 0)$ enters an electric field $\vec{E} = -E_0\hat{i}$ (E_0 = constant > 0) at $t = 0$. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is –

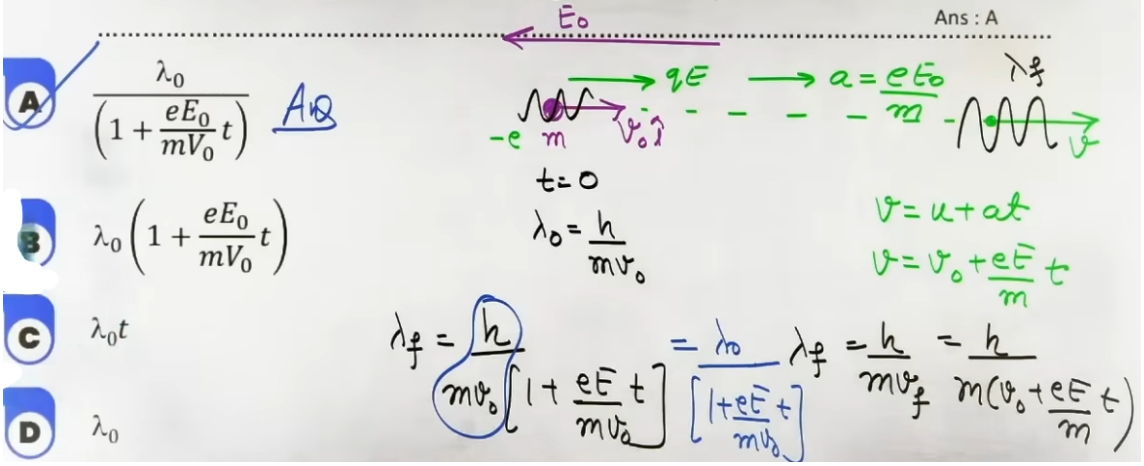
Ans : A

A. $\frac{\lambda_0}{\left(1 + \frac{eE_0}{mV_0}t\right)}$ Ans

B. $\lambda_0\left(1 + \frac{eE_0}{mV_0}t\right)$

C. $\lambda_0 t$

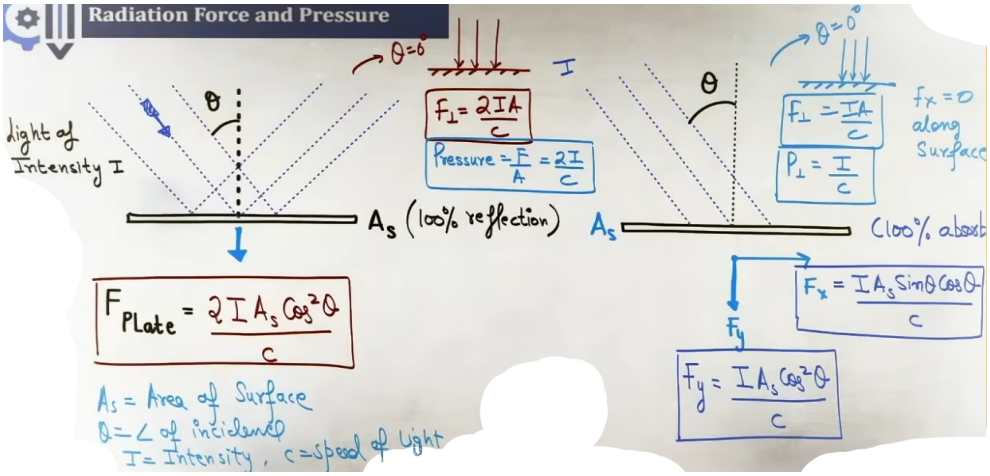
D. λ_0



RADIATION FORCE AND RADIATION PRESSURE

Physical Mechanism:

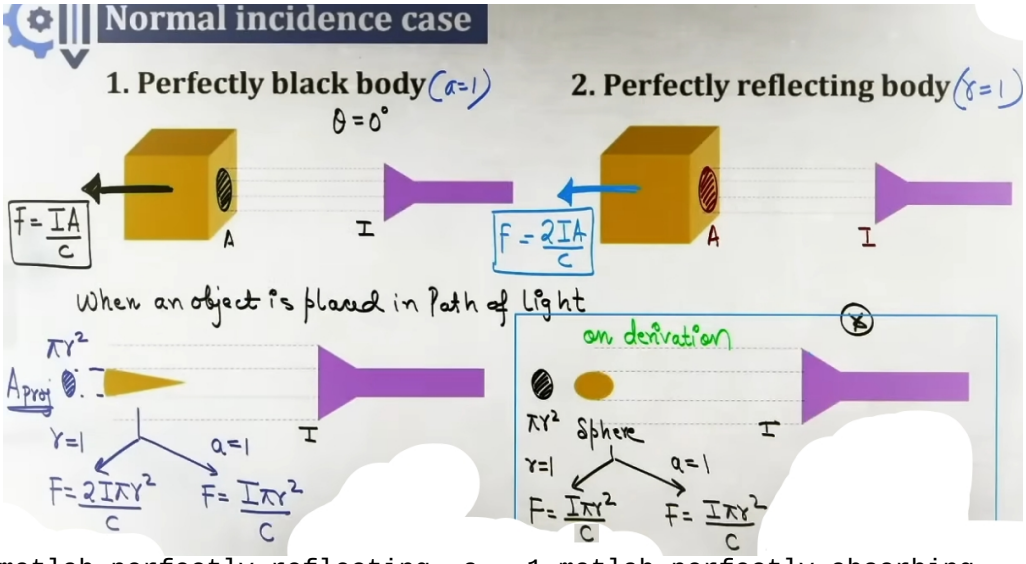
Photons carry linear momentum p = E/c. When light strikes a surface, momentum transfer occurs, generating radiation force F_rad = Δp/Δt and radiation pressure P_rad = F_rad/Area.



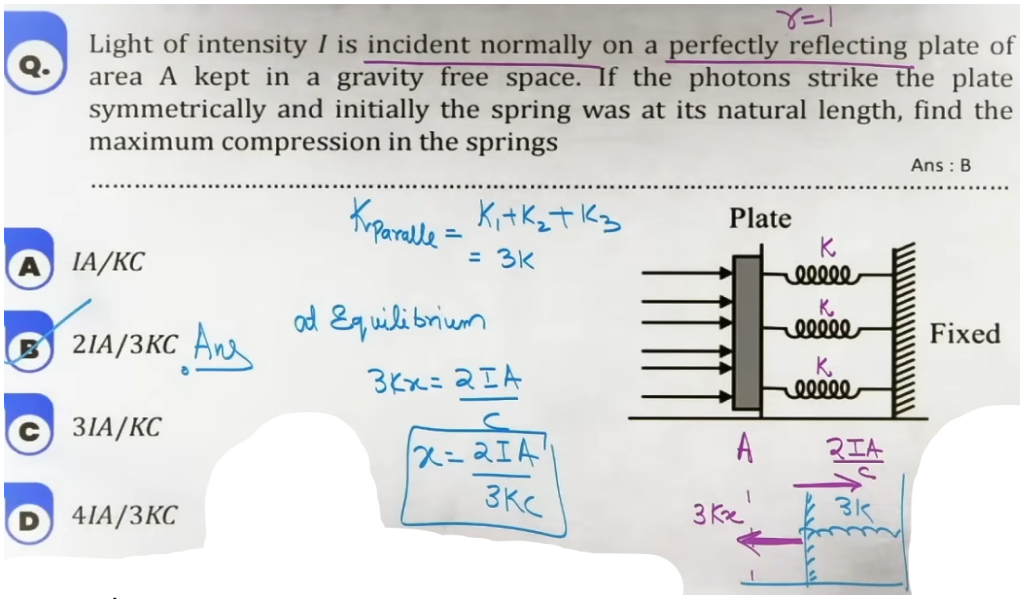
Case 1: Inclined Incidence (Angle θ with Surface Normal, Intensity I, Surface Area A)
- 100% Absorbing Surface:
  Perpendicular Force: F_perp = (IAcos²θ)/c
  Parallel/Tangential Force: F_parallel = (IAsinθcosθ)/ c
  Radiation Pressure: P_rad = F_perp / A = (Icos²θ)/c
- 100% Reflecting Surface:
  Perpendicular Force: F_perp = (2IAcos²θ)/c
  Parallel/Tangential Force: F_parallel = 0
  Radiation Pressure: P_rad = F_perp/A = (2Icos²θ)/c

Case 2: Radiation Force on 3D/Curved Objects (Beam of Intensity I along symmetry axis)
- Cone or Cylinder (Base Radius R, Base Area A_proj = πR²):
  100% Absorbing: F = (IπR²)/c
  100% Reflecting: F = (2IπR²)/ c
- Sphere of Radius R (Cross-sectional Area A_proj = πR²):
  CRUCIAL DERIVATION RESULT:
  100% Absorbing Sphere: F = (IπR²)/c
  100% Reflecting Sphere: F = (IπR²)/c

(SPECIAL NOTE: Due to integration of force components over the spherical curved surface, the factor of 2 cancels out for a reflecting sphere. Both 100% absorbing and 100% reflecting spheres experience the EXACT SAME FORCE F = IπR²/c).



r = 1 matlab perfectly reflecting, a = 1 matlab perfectly absorbing.



sphere exception

FORMULA SHEET

- Einstein's Photoelectric Equation:
  $h * \nu = \phi + K_{max} = h * \nu_0 + e * V_s$
  $K_{max} = e * V_s = (1/2) * m * (v_{max})^2$

- Stopping Potential:
  $V_s = (h/e)*\nu - (\phi / e) = (hc/e)*(1/\lambda) - (\phi / e)$

- Photon Emission Rate (Photons/sec emitted by source of Power P):
  $n_p = N_p / t = P / E_p = (P * \lambda) / (h * c)$

- De Broglie Wavelength Formulas:
  General: $\lambda = h/p = h/(mv)$
  Kinetic Energy Form: $\lambda = h/\sqrt{(2mK)}$
  Voltage Accelerated Charge Form: $\lambda = h/\sqrt{(2mqV)}$

- Accelerated Particle Shortcuts (V in Volts, λ in Ångströms):
  Electron : $\lambda_e = 12.27 / \sqrt{V}$ Å
  Proton : $\lambda_p = 0.286 / \sqrt{V}$ Å
  Deuteron : $\lambda_d = 0.202 / \sqrt{V}$ Å
  Alpha Particle: $\lambda_\alpha = 0.101 / \sqrt{V}$ Å

- De Broglie Application Formulas:
  Thermal Gas: $\lambda_{thermal} = h / \sqrt{(3 * m * k_B * T)}$
  Magnetic Field B (Radius R): $\lambda = h/(qBR)$
  Perpendicular Fused Collision: $\lambda_{net} = (\lambda_1 * \lambda_2) / \sqrt{(\lambda_1^2 + \lambda_2^2)}$
  Explosion at Rest: $\lambda_1 = \lambda_2$

- Radiation Force & Pressure Formulas (Normal Incidence, Beam Power P = I*A):
  100% Absorbing: $F = (I*A)/c$; $P_{rad} = I / c$
  100% Reflecting (Flat Surface): $F = (2IA) / c$; $P_{rad} = 2I/c$
  100% Reflecting Sphere (Radius R): $F = (I\pi R^2)/c$

CONCEPT CONNECTIONS

- Connection to Kinetic Theory of Gases (KTG):
  Translational kinetic energy $K = (3/2) k_B * T$ bridges thermal thermodynamics to quantum matter waves $\lambda = h/\sqrt{(3m k_B T)}$.

Comparison Table: Radiation Force on Geometries :

Geometry	100% Absorbing Surface	100% Reflecting Surface
Flat Plate (Normal θ=0)	$(IA)/c$	$(2IA)/c$
Flat Plate (Angle θ)	$(IA\cos^2\theta)/c$	$(2IA\cos^2\theta)/c$
Cylinder / Cone	$(I\pi R^2)/c$	$(2I\pi R^2)/c$
Sphere	$(I\pi R^2)/c$	$(I\pi R^2)/c$ (Special Exception!)

JEE ADVANCED INSIGHTS

1. Reflecting Sphere Force Equivalence:
A flat 100% reflecting plate experiences force $F = 2IA/c$. But a 100% reflecting SPHERE experiences force $F = I\pi R^2/c$ (same as an absorbing sphere!). Do not insert the factor of 2 for a sphere in numericals.

2. Identical Wavelength Fragile Decays:
When a stationary particle breaks into two fragments of masses m_1 and m_2, both fragments MUST have identical De Broglie wavelengths ($\lambda_1 = \lambda_2$), regardless of mass ratio, because linear momentum is conserved equally in opposite directions.

HIGH-YIELD REVISION SHEET
MUST REMEMBER:

Key Formulas:
- Einstein's Photoelectric Equation: $h * \nu = \phi + K_{max} = \phi + e * V_s$
- Photon Emission Rate: $n_p = P / E_p = (P * \lambda) / (h * c)$
- De Broglie Wavelength: $\lambda = h / p = h / (m * v) = h / \sqrt{(2 * m * K)}$
- Accelerated Charges: $\lambda = h / \sqrt{(2 * m * q * V)}$
- Electron Shortcut: $\lambda_e = 12.27 / \sqrt{V}$ Å
- Proton Shortcut: $\lambda_p = 0.286 / \sqrt{V}$ Å
- Alpha Shortcut: $\lambda_\alpha = 0.101 / \sqrt{V}$ Å
- Thermal Gas Wavelength: $\lambda = h / \sqrt{(3 * m * k_B * T)}$
- Orthogonal Fusion Collision: $\lambda_{net} = (\lambda_1 * \lambda_2) / \sqrt{(\lambda_1^2 + \lambda_2^2)}$
- Explosion at Rest: $\lambda_1 = \lambda_2$
- Radiation Force (Flat Absorbing): $F = (I * A) / c$
- Radiation Force (Flat Reflecting): $F = (2 * I * A) / c$
- Radiation Force (Sphere Reflecting or Absorbing): $F = (I * \pi * R^2) / c$

ATOMS

Overview of Atomic Models

Thomson's Model (Plum Pudding / Watermelon Model):
- An atom is envisioned as a uniform sphere of positive charge of radius approximately 10^{-10} m in which negative electrons are embedded throughout like seeds in a watermelon.
  Failure: Failed to explain large-angle scattering of alpha particles and discrete atomic spectral lines.

Rutherford's Alpha Particle Scattering Experiment:
- Experimental Setup: High-energy alpha particles (He^{2+}, mass = 4 amu, charge = +2e) emitted from a radioactive bismuth/polonium source are collimated into a narrow beam and directed toward a thin gold foil (thickness ≈ 2.1×10^{-7} m).
  Why Gold Foil was Chosen:
  1. Gold is extremely malleable and can be hammered into an ultrathin foil (ensures a single scattering event per alpha particle).
  2. Gold has a high atomic number (Z = 79), providing a massive, heavy nucleus that remains stationary during interaction, so all observable momentum change occurs in the lighter alpha particle.

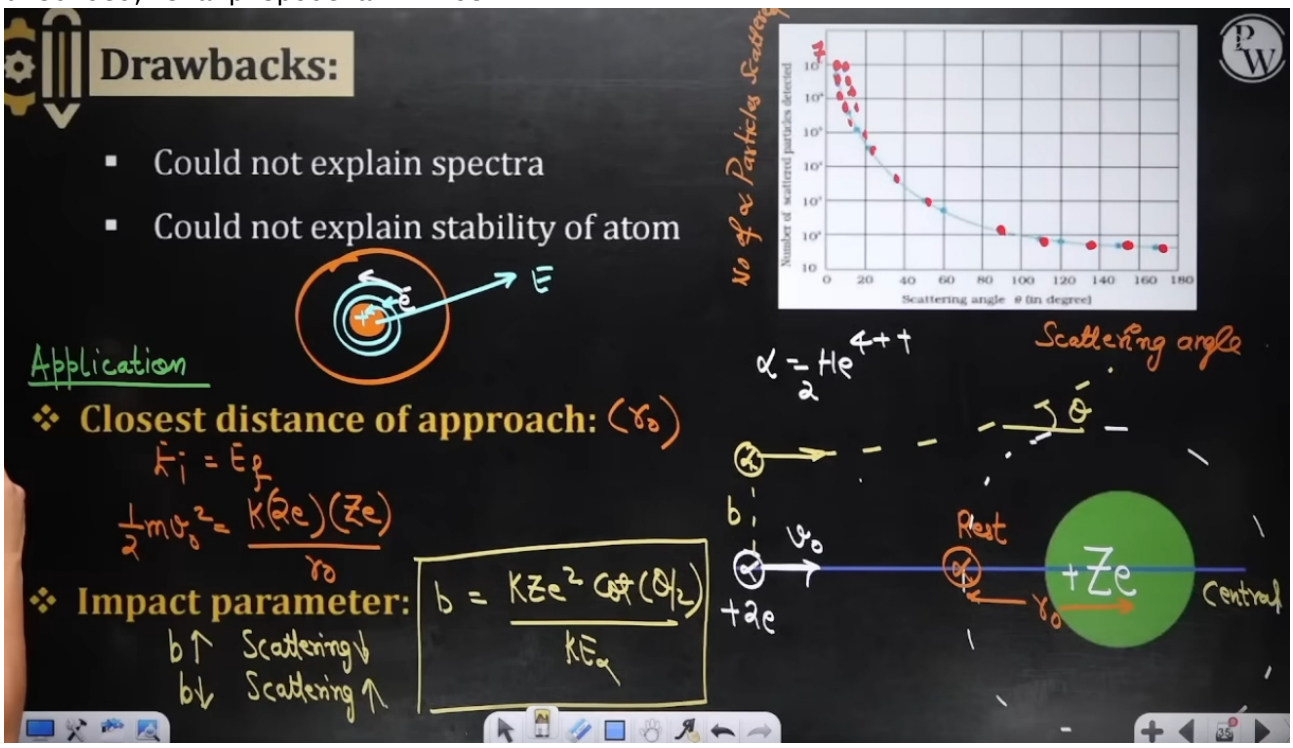
Observations, Graphs, and Drawbacks of Rutherford's Model

Experimental Observations:
  1. Overwhelming majority (> 99.9%) of alpha particles pass straight through the gold foil without any measurable deflection.
  2. A small fraction (approx. 1 in 8000 or 0.14%) is deflected by small angles ($\theta > 1°$).
  3. An extremely tiny fraction (less than 1 in 8000) is deflected through large angles ($\theta > 90°$), with very few rebounding almost straight back ($\theta \approx 180°$).

Conclusions / Interpretations:
  1. Atom is mostly empty space.
  2. The entire positive charge and almost the entire mass of the atom are concentrated in a tiny, dense core called the nucleus.
  3. The nucleus occupies a minute fraction of the total atomic volume (nucleus radius ~ 10^{-15} m vs atom radius ~ 10^{-10} m).

Drawbacks of Rutherford's Model:
  1. Inability to Explain Atomic Orbital Stability (Collapse of Atom): According to classical electromagnetic theory, an accelerating charge (electron revolving in a circular orbit) must continuously radiate energy in the form of electromagnetic waves. Losing energy causes its orbital radius to decrease continuously, producing a spiral trajectory that ends with the electron collapsing into the nucleus within 10^{-10} seconds.

2. Inability to Explain Discrete Line Spectra: Continuous energy loss predicts a continuous spectrum containing all frequencies. Experimentally, atoms emit discrete, sharp spectral lines.



Distance of Closest Approach (r_0) and Impact Parameter (b)

Distance of Closest Approach (r_0):

- Definition: The minimum distance from the center of the target nucleus to which an alpha particle traveling directly along the central axis (head-on collision) can approach before coming to instantaneous rest and reversing its motion.

Derivation via Conservation of Mechanical Energy:

$r_0 = (2\ k\ Z\ e^2) / K_\alpha = (4\ k\ Z\ e^2) / (m_\alpha\ v_0^2)$

where $k = 1 / (4\pi\ \varepsilon_0) \approx 9 \times 10^9$ N m^2/C^2, Z is target atomic number, $e = 1.6 \times 10^{-19}$ C, m_α is alpha particle mass.

Impact Parameter (b): # not so imp :P

- Definition: The perpendicular distance between the initial velocity vector of the alpha particle (when far from the nucleus) and the parallel central axis passing through the nucleus.

Mathematical Formula:

$b = [Ze^2\cot(\theta/2)]/[4\pi\varepsilon_0 K_\alpha] = [kZe^2\cot(\theta/2)]/[(1/2)m_\alpha v_0^2]$

- Key Relationships:

1. Large Impact Parameter b: Alpha particle passes far from the nucleus, experiencing weak Coulomb repulsion -> Small scattering angle θ.
2. Small Impact Parameter b: Alpha particle passes close to the nucleus, experiencing strong Coulomb repulsion -> Large scattering angle θ.
3. Zero Impact Parameter (b = 0): Head-on collision along the central axis -> Maximum scattering angle θ = 180° (alpha particle bounces straight back).

Bohr's Postulates for Hydrogen and Hydrogen-like Atoms

Applicability: Valid strictly for single-electron species (such as H, He^+, Li^{2+}, Be^{3+}) with atomic number Z and nuclear charge +Ze.

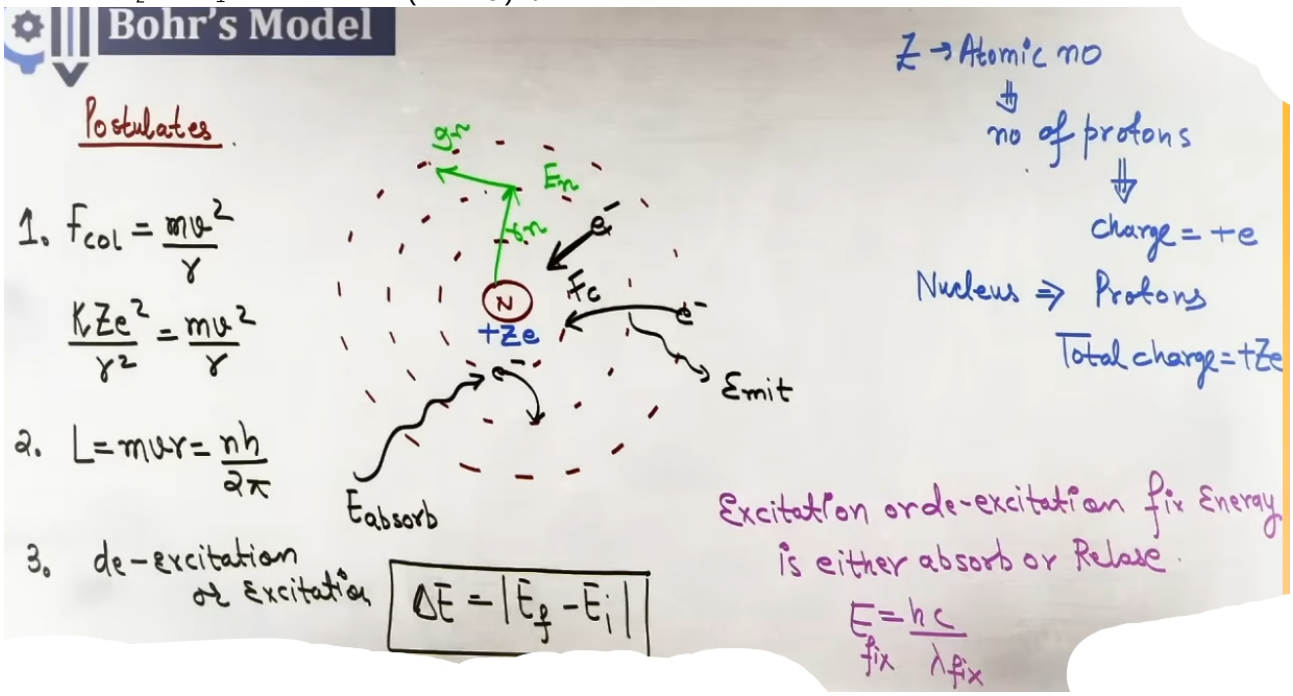
- Postulate 1 (Stationary Orbits and Centripetal Force Balance):

An electron revolves around the positive nucleus in specific non-radiating circular orbits. The electrostatic Coulomb attraction between the nucleus (+Ze) and the electron (-e) provides the necessary centripetal force:

$F_{centripetal} = F_{electrostatic}$

$(m * v^2) / r = (1 / (4\pi\ \varepsilon_0)) * (Z * e^2) / r^2 = (k * Z * e^2) / r^2$

- Postulate 2 (Quantization of Orbital Angular Momentum):
  An electron can revolve only in those non-classical stationary orbits where its orbital angular momentum L is an integral multiple of h / (2π):
  L = mvr = (n)*(h/2π) where n = 1, 2, 3... (principal quantum number).
- Postulate 3 (Transition and Energy Quantum Condition):
  An electron in a stationary orbit does not radiate energy. Electromagnetic radiation is emitted or absorbed only when an electron jumps from one permitted orbit to another:
  $\Delta E = E_2 - E_1 = h * \nu = (h * c) / \lambda$

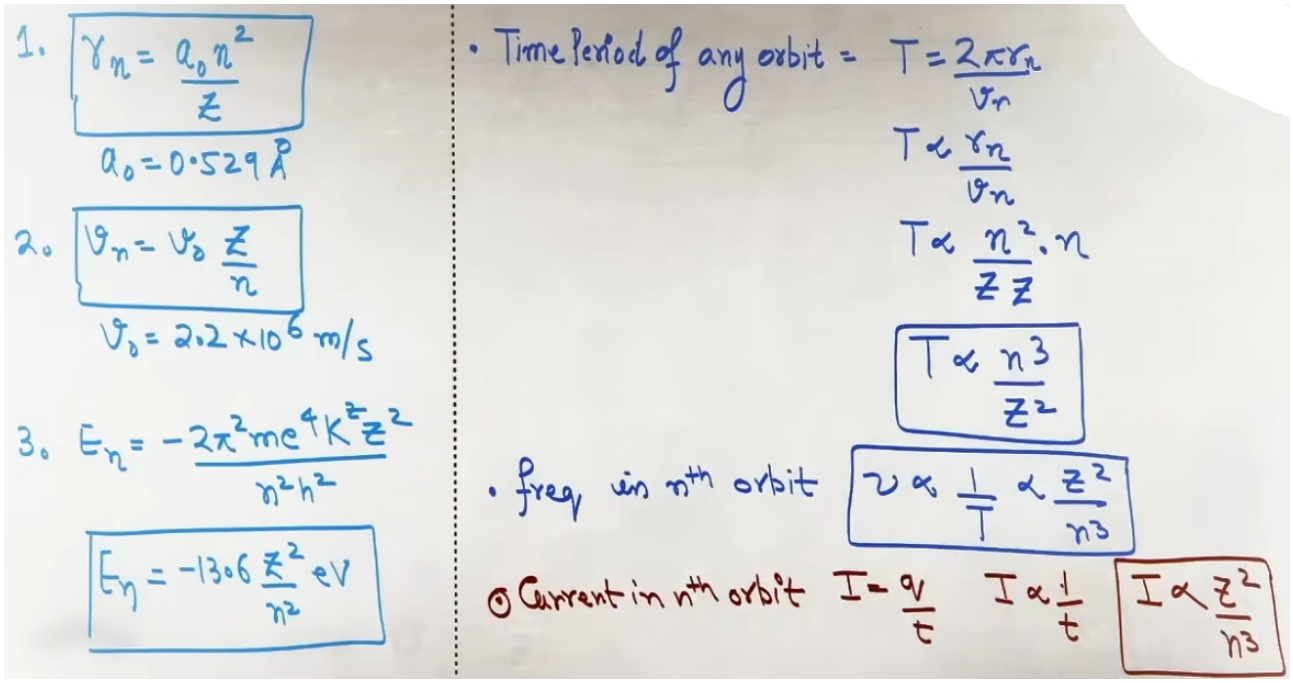


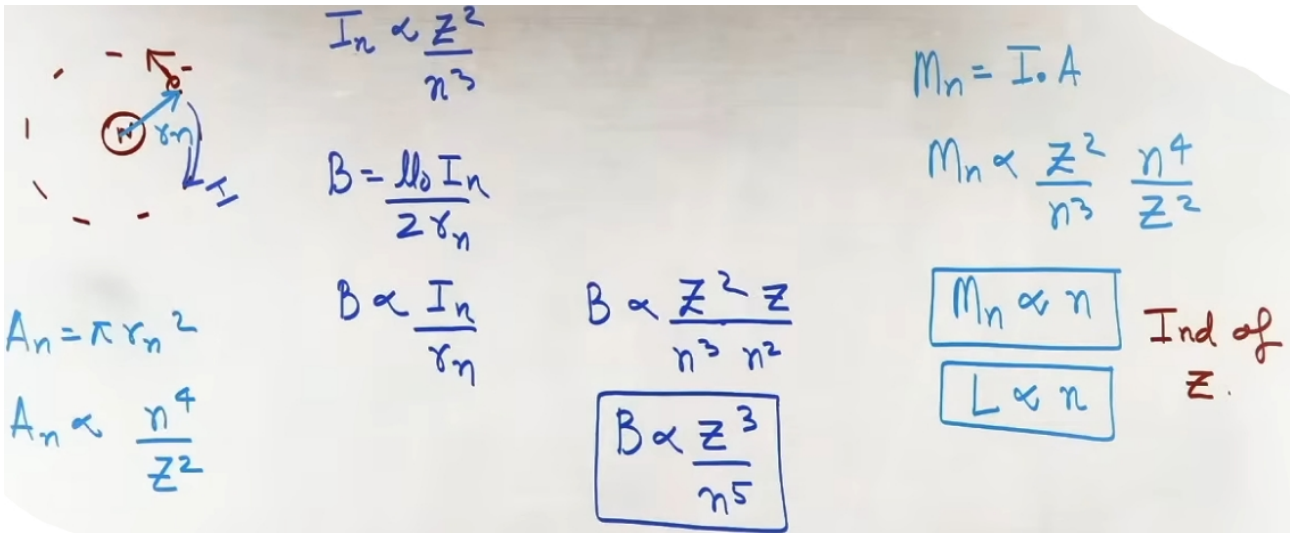
Derived Formulas for n-th Orbit Parameters
Solving Postulates 1 and 2 simultaneously yields all physical parameters for an electron in the n-th orbit:
- 1. Radius of n-th Orbit (r_n):
  From L = m v r = n h / (2π) => v = n h / (2π m r).
  Substituting v into (m v²) / r = (k Z e²) / r²:
  $r_n = [n^2 h^2 \varepsilon_0] / [\pi m Z e^2] =$ **$r_0*(n^2/Z) = 0.529*(n^2/Z)$ Å**
  where $r_0 = a_0 = 0.529$ Å = 0.0529 nm is the Bohr radius (ground state radius of hydrogen, n=1, Z=1).
  Proportionality: $r_n \propto n^2 / Z$.

- 2. Velocity of Electron in n-th Orbit (v_n):
  $v_n = [Z e^2] / [2 \varepsilon_0 n h] =$ **$v_0*(Z/n) = 2.18 \times 10^6*(Z/n)$ m/s**
  where $v_0 = 2.18 \times 10^6$ m/s is the orbital velocity in the ground state of hydrogen ($v_0 \approx c / 137$).
  Proportionality: $v_n \propto Z / n$.

- 3. Energies of Electron in n-th Orbit (E_n, K_n, U_n):
  Kinetic Energy (K_n): $K_n = (1/2)m(v_n^2) = (kZe^2)/(2r_n) =$ **$+13.6*(Z^2/n^2)$ eV.**
  Potential Energy (U_n): $U_n = - (k Z e^2) / r_n =$ **$-27.2*(Z^2/n^2)$ eV.**
  Total Energy (E_n = K_n + U_n):
  $E_n = -(kZe^2)/(2r_n) = -[mZ^2 e^4 k^2 2\pi^2] / [n^2h^2] =$ **$-13.6*(Z^2/n^2)$ eV.**
  Fundamental Energy Inter-relationships:
  K_n = - E_n
  *U_n = 2 * E_n*
  E_n = U_n / 2 = - K_n
  Proportionality: $E_n \propto Z^2 / n^2$, $K_n \propto Z^2 / n^2$, $U_n \propto Z^2 / n^2$.

- 4. Time Period of Revolution (T_n):
  $T_n = (2 \pi r_n) / v_n \propto (n^2 / Z) / (Z / n) = n^3 / Z^2$.
  Formula: **$T_n = T_0 * (n^3 / Z^2)$ where $T_0 \approx 1.52 \times 10^{-16}$ s.**





CRUCIAL OBSERVATION: Magnetic dipole moment M_n is directly proportional to principal quantum number n and is completely INDEPENDENT of atomic number Z!

INTUITION AND CONCEPTUAL UNDERSTANDING

- Impact Parameter and Deflection Geometry:
  Think of a dart thrown toward a round target.
  A dart aiming right at the bullseye (b = 0) hits dead center and bounces straight back (θ = 180°).
  A dart aiming far to the side (large b) barely grazes the target's force field, deflecting by a tiny angle θ.

- Energy Relationships Intuition (The Negative Energy Well):
  Total energy E_n is negative (-13.6 eV for H ground state) because the electron is trapped in a bound state inside the nucleus's attractive potential well.
  To free the electron (move it to n = ∞ where E = 0), an external agency must supply energy equal to +13.6 eV (Ionization Energy).
  Potential energy U_n is always twice as negative as total energy E_n (U_n = 2 E_n), while kinetic energy K_n is always positive and equal in magnitude to total energy (K_n = -E_n).

FORMULA SHEET

- Distance of Closest Approach (Head-on collision b = 0):
  $r_0 = (2\ k\ Z\ e^2) / K_\alpha = (4\ k\ Z\ e^2) / (m_\alpha\ v_0^2)$
  Units: meters (m) or femtometers (1 fm = 10^{-15} m).

- Impact Parameter:
  $b = [\ k\ Z\ e^2\ \cot(\theta / 2)\ ] / K_\alpha$
  Scattering Angle Relation: $N(\theta) \propto 1 / \sin^4(\theta / 2)$

- Bohr Angular Momentum Quantization:
  $L = m * v * r = (n * h) / (2\pi)$

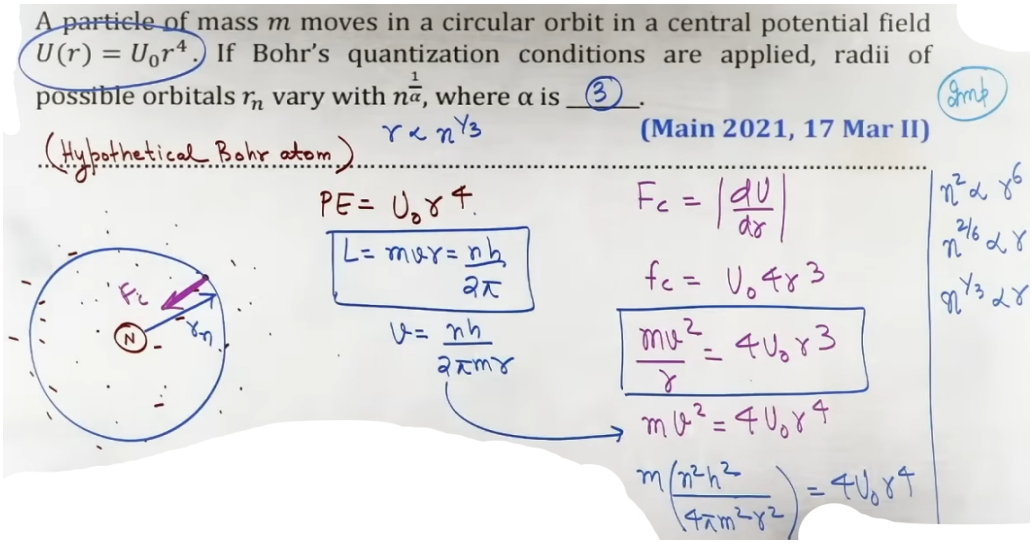
- Bohr Orbital Parameters for n-th Orbit:
  Radius : $r_n = 0.529 * (n^2 / Z)$ Å
  Velocity : $v_n = 2.18 \times 10^6 * (Z / n)$ m/s
  Total Energy : $E_n = -13.6 * (Z^2 / n^2)$ eV
  Kinetic Energy : $K_n = +13.6 * (Z^2 / n^2)$ eV
  Potential Energy: $U_n = -27.2 * (Z^2 / n^2)$ eV

- Scaling Proportionalities with Quantum Number n and Atomic Number Z:
  Radius : $r_n \propto n^2 / Z$
  Velocity : $v_n \propto Z / n$
  Total / Kinetic / Potential Energy: $E_n, K_n, U_n \propto Z^2 / n^2$
  Time Period : $T_n \propto n^3 / Z^2$
  Frequency of Revolution : $f_n \propto Z^2 / n^3$
  Orbital Current : $I_n \propto Z^2 / n^3$
  Magnetic Field at Center : $B_n \propto Z^3 / n^5$
  Magnetic Dipole Moment : $M_n \propto n$ (Independent of Z)

EXAMPLES

Hypothetical Bohr Atom with Non-Standard Force (JEE Advanced Pattern)
Problem: Suppose an electron moves in a central potential field where potential energy is $U(r) = C * r^4$. Assuming Bohr angular momentum quantization holds, find how orbit radius r depends on principal quantum number n.
Solution:
1. Force $F(r) = -dU / dr = -d(C * r^4) / dr = -4\ C\ r^3$.
2. Centripetal force balance:
   $m * v^2 / r = 4\ C\ r^3 \Rightarrow m * v^2 = 4\ C\ r^4 \Rightarrow v^2 = (4\ C\ r^4) / m$.
3. Bohr angular momentum quantization:
   $m * v * r = (n * h) / (2\pi) \Rightarrow v = (n * h) / (2\ \pi\ m\ r)$.
4. Square velocity expression and equate:
   $v^2 = (n^2\ h^2) / (4\ \pi^2\ m^2\ r^2)$
   $(4\ C\ r^4) / m = (n^2\ h^2) / (4\ \pi^2\ m^2\ r^2)$
   $r^6 = (n^2\ h^2) / (16\ \pi^2\ m\ C)$
   $r \propto n^{(2/6)} = n^{(1/3)}$.

Therefore, orbit radius r is directly proportional to $n^{(1/3)}$.



- Hypothetical Bohr Atoms (Modified Force or Potential Energy Laws):
  JEE Advanced questions frequently construct hypothetical atomic systems where the potential energy U(r) or central repulsive/attractive force F(r) differs from the standard $1/r^2$ Coulomb law.
  In such problems, standard formulas ($r \propto n^2$, $v \propto 1/n$, $E \propto 1/n^2$) DO NOT APPLY and will give incorrect answers.

- Universal Four-Step Solution Framework for Hypothetical Bohr Atoms:
  Step 1: Obtain the central attractive force F(r) from the given potential energy U(r) using $F(r) = -dU / dr$.
  Step 2: Balance the centripetal force with the central force: $mv^2 / r = F(r)$.
  Step 3: Apply Bohr's angular momentum quantization postulate (assumed to remain valid): $m * v * r = (n * h) / (2\pi) \Rightarrow v = (n * h) / (2\pi m r)$.
  Step 4: Substitute v from Step 3 into the force balance equation from Step 2 to solve for orbit radius r as a function of quantum number n.
  Step 5: Calculate total energy E(n) by summing kinetic energy $K = (1/2) m v^2$ and given potential energy U(r).

JEE ADVANCED INSIGHTS

1. Magnetic Moment Independence from Z:
The orbital magnetic dipole moment $M_n = I_n * A_n$ is directly proportional to principal quantum number n ($M_n \propto n$) and is completely independent of atomic number Z. For any hydrogen-like ion in the ground state (n = 1), M_1 is identical (equal to one Bohr Magneton $\mu_B = e h / (4\pi m)$).

2. Distance of Closest Approach as Nuclear Size Upper Bound:
The distance of closest approach r_0 provides an upper boundary for the radius of the nucleus ($R_{nucleus} < r_0$). Alpha particles with kinetic energy up to ~ 5-10 MeV cannot penetrate the nucleus, so nuclear forces do not alter classical Coulomb scattering at these energies.

HIGH-YIELD REVISION SHEET

Key Formulas:
- Distance of Closest Approach: $r_0 = (2 k Z e^2) / K_\alpha = (4 k Z e^2) / (m_\alpha v_0^2)$
- Impact Parameter: $b = [k Z e^2 \cot(\theta / 2)] / K_\alpha$
- Angular Momentum Quantization: $m * v * r = (n * h) / (2\pi)$
- Bohr Radius: $r_n = 0.529 * (n^2 / Z)$ Å
- Orbital Velocity: $v_n = 2.18 \times 10^6 * (Z / n)$ m/s
- Total Energy: $E_n = -13.6 * (Z^2 / n^2)$ eV
- Kinetic Energy: $K_n = +13.6 * (Z^2 / n^2)$ eV $= - E_n$
- Potential Energy: $U_n = -27.2 * (Z^2 / n^2)$ eV $= 2 E_n$

Key Observations:
1. Rutherford scattering formula: $N(\theta) \propto 1 / \sin^4(\theta / 2)$.
2. Large b implies small scattering angle θ; b = 0 corresponds to θ = 180°.
3. Rutherford's model failed due to classical collapse of atom (orbital decay) and inability to explain discrete line spectra.
4. Magnetic dipole moment $M_n \propto n$ is independent of target atomic number Z.
5. Magnetic field at nucleus scales steeply as $B_n \propto Z^3 / n^5$.

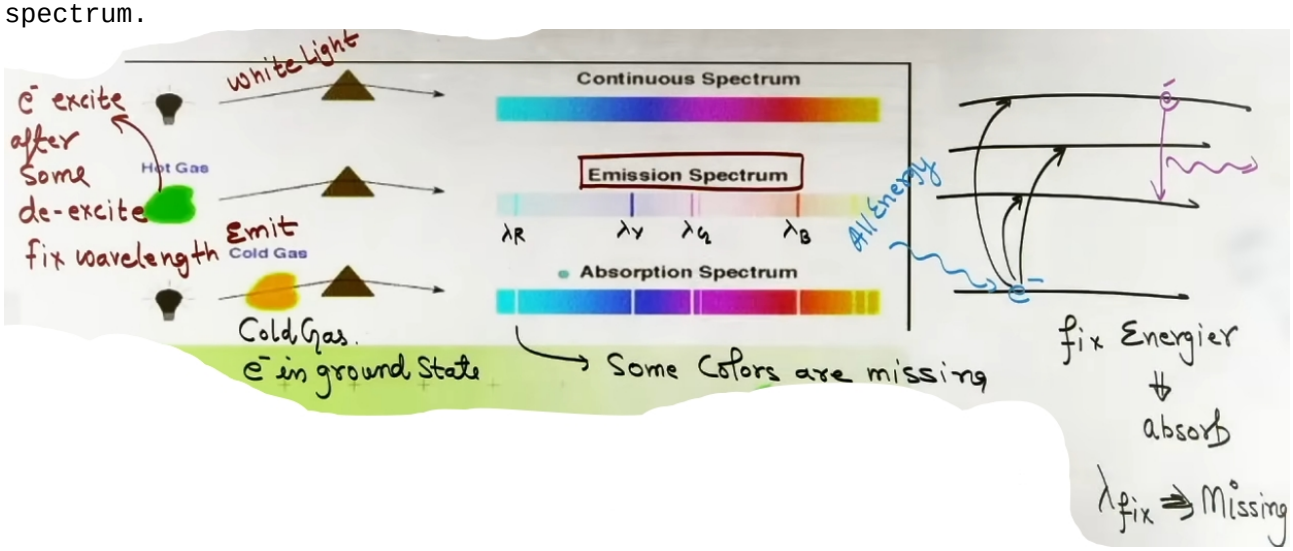
ATOMS - SPECTRA, RYDBERG FORMULA, AND BOHR LIMITATIONS

Continuous, Emission, and Absorption Spectra
Continuous Spectrum:
- A continuous spectrum is produced when white light (such as light from an incandescent bulb or hot filament) passes through a dispersing prism or grating.
- It contains an unbroken band of all wavelengths and colors from violet to red (VIBGYOR) without any dark gaps or missing frequencies.

Emission Spectrum:
- An emission spectrum is obtained when a gas at low pressure is thermally or electrically excited.
- When excited electrons in the gas atoms de-excite from higher energy levels to lower energy levels, they emit electromagnetic radiation of specific, quantized wavelengths corresponding to exact differences between energy states.
- An emission spectrum consists of bright, distinct colored lines separated by dark spaces against a black background.
- Because every element possesses a unique set of quantized energy levels, the emission spectrum serves as a unique characteristic "fingerprint" of that element.

Absorption Spectrum:
- An absorption spectrum is produced when continuous white light containing all wavelengths is passed through a cold gas whose atoms are in their ground state.
- The electrons in the ground state absorb only those specific photon energies that match the exact energy differences needed to jump to higher excited states.
- The transmitted light, when dispersed, shows dark lines or missing bands at those exact absorbed wavelengths against an otherwise continuous bright background.

Complementary Relation Between Spectra:
- The dark absorption lines of an element appear at the exact same wavelengths as its bright emission lines.
- If an emission spectrum and an absorption spectrum of the same element are superimposed, they completely fill each other to reproduce the full continuous spectrum.



Rydberg Formula and Spectral Series

Rydberg Formula for Emitted Wavelength:
- When an electron in a hydrogen-like single-electron atom transitions from an upper energy state n_2 to a lower energy state n_1 (where $n_2 > n_1$), the wavelength λ of the emitted photon is given by the Rydberg formula:

$$\mathbf{1 / \lambda = R * Z^2 * [(1 / n_1^2) - (1 / n_2^2)]}$$

where:
R = Rydberg constant $\approx 1.097 \times 10^7\ m^{-1}$.
Z = Atomic number of the hydrogen-like species ($Z = 1$ for H, $Z = 2$ for He^+).
n_1 = Principal quantum number of the lower energy level to which the electron de-excites.
n_2 = Principal quantum number of the higher energy level from which the electron originates.

Energy Equivalent of Rydberg Constant:
- The energy associated with the Rydberg constant is $R * h * c \approx 13.6\ eV$.
- Photon energy in transition:

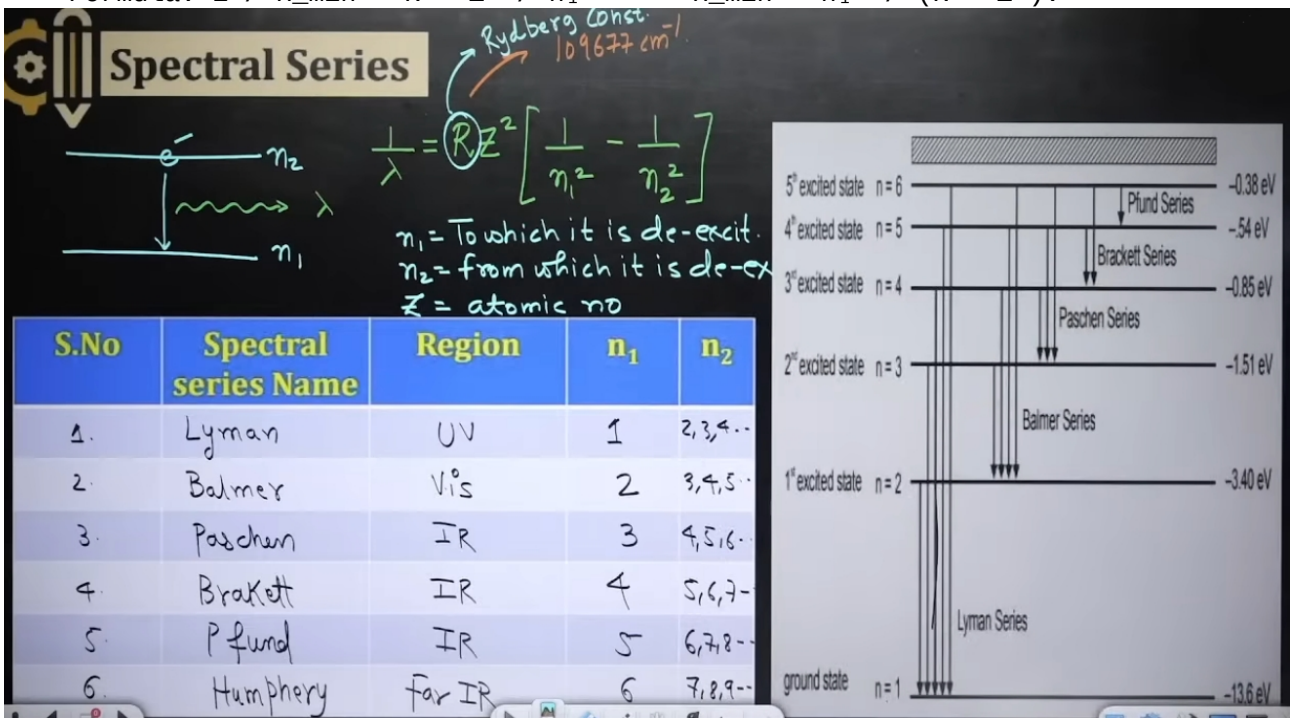
$$\mathbf{\Delta E = E_{n2} - E_{n1} = 13.6 * Z^2 * [(1 / n_1^2) - (1 / n_2^2)]\ eV.}$$

Spectral Series of Hydrogen Atom (Z = 1):

1. Lyman Series:
   - De-excitation to $n_1 = 1$ from upper states $n_2 = 2, 3, 4, 5....$
   - Spectral Region: Ultraviolet (UV) region.
2. Balmer Series:
   - De-excitation to $n_1 = 2$ from upper states $n_2 = 3, 4, 5, 6....$
   - Spectral Region: Visible light region.
   - **Note:** Balmer series is the only spectral series of hydrogen that falls in the human visible spectrum.
3. Paschen Series:
   - De-excitation to $n_1 = 3$ from upper states $n_2 = 4, 5, 6, 7....$
   - Spectral Region: Infrared (IR) region.
4. Brackett Series:
   - De-excitation to $n_1 = 4$ from upper states $n_2 = 5, 6, 7, 8....$
   - Spectral Region: Infrared (IR) region.
5. Pfund Series:
   - De-excitation to $n_1 = 5$ from upper states $n_2 = 6, 7, 8, 9....$
   - Spectral Region: Infrared (IR) region.
6. Humphrey Series:
   - De-excitation to $n_1 = 6$ from upper states $n_2 = 7, 8, 9, 10....$
   - Spectral Region: Far Infrared region.

Series Limits and Extreme Wavelengths:

- Maximum Wavelength (λ_max) of a Series:
  Occurs for the minimum energy transition within the series, which takes place from the immediate adjacent level $n_2 = n_1 + 1$.
  Formula: $1 / \lambda_{max} = R * Z^2 * [(1 / n_1^2) - (1 / (n_1 + 1)^2)]$.
- Minimum Wavelength / Series Limit (λ_min) of a Series:
  Occurs for the maximum energy transition within the series, which takes place from infinity $n_2 = \infty$.
  Formula: $1 / \lambda_{min} = R * Z^2 / n_1^2 \Rightarrow \lambda_{min} = n_1^2 / (R * Z^2)$.

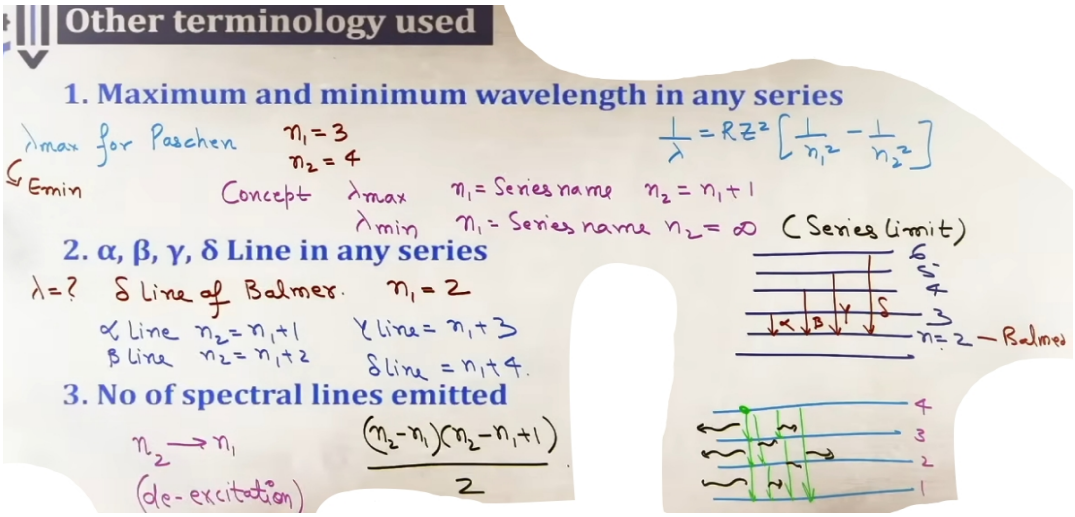


Spectral Line Designations (α, β, γ, δ) and Total Number of Emitted Lines

Greek Letter Designations within a Series:

- Within any spectral series characterized by lower state n_1, individual emission lines are designated based on the upper state n_2:
  1. Alpha (α) Line: Transition from $n_2 = n_1 + 1$ to n_1 (first line, lowest energy, longest wavelength in series).
  2. Beta (β) Line: Transition from $n_2 = n_1 + 2$ to n_1 (second line).
  3. Gamma (γ) Line: Transition from $n_2 = n_1 + 3$ to n_1 (third line).
  4. Delta (δ) Line: Transition from $n_2 = n_1 + 4$ to n_1 (fourth line).

Example for Balmer Series ($n_1 = 2$):
- H_α line: Transition from $n_2 = 3$ to $n_1 = 2$.
- H_β line: Transition from $n_2 = 4$ to $n_1 = 2$.
- H_γ line: Transition from $n_2 = 5$ to $n_1 = 2$.
- H_δ line: Transition from $n_2 = 6$ to $n_1 = 2$.



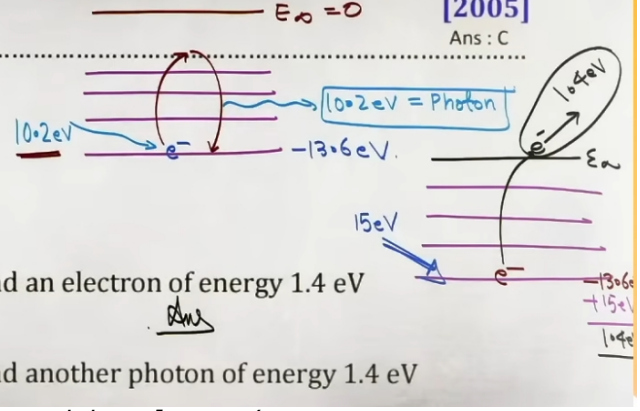
<u>Total Number of Emitted Spectral Lines:</u>
- When a gas sample containing a large number of excited hydrogen-like atoms de-excites from an initial higher state N_2 to a lower state N_1:
  Total Number of Spectral Lines = $(N_2 - N_1) * (N_2 - N_1 + 1) / 2$.
- Special Case (De-excitation to Ground State $N_1 = 1$):
  When electrons de-excite from state n to the ground state n = 1:
  Total Number of Spectral Lines = $n * (n - 1) / 2$.

Limitations of Bohr Model and Hypothetical Bohr Atoms

- This model could not explain the fine structure of spectral lines
- There was splitting of spectral lines in presence of magnetic field and electric field which could not be explained by Bohr's model.
- This model is only valid for single electron atoms (1e⁻ Should be there)
- This model is based on circular orbits of electron but in reality there are no circular orbits.
- This model could not explain the intensity of spectral lines
- This model only considered particle nature of electron.

Q. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector? [2005]

Ans : C



A 2 photons of energy 10.2 eV

B 2 photons of energy 1.4 eV

C One photon of energy 10.2 eV and an electron of energy 1.4 eV

D One photon of energy 10.2 eV and another photon of energy 1.4 eV

hydrogen k har shell ki energy yaad kar lena :(

FORMULA SHEET

- Rydberg Formula:
  $1 / \lambda = R * Z^2 * [(1 / n_1^2) - (1 / n_2^2)]$
  $R \approx 1.097 \times 10^7 \ m^{-1}$, $R * h * c \approx 13.6$ eV

- Series Extreme Wavelengths:
  Maximum Wavelength: $1 / \lambda_{max} = R * Z^2 * [(1 / n_1^2) - (1 / (n_1 + 1)^2)]$
  Minimum Wavelength (Series Limit): $\lambda_{min} = n_1^2 / (R * Z^2)$

- Spectral Line Count Formulas:
  Transition between N_2 and N_1: Number of lines = $(N_2 - N_1) * (N_2 - N_1 + 1) / 2$
  Transition from n to Ground State ($n_1 = 1$): Number of lines = $n * (n - 1) / 2$

- Hypothetical Bohr Atom General Equations:
  $F(r) = -dU / dr$
  $m * v^2 / r = F(r)$
  $m * v * r = (n * h) / (2\pi)$
  $E_{total} = K + U = (1/2) m v^2 + U(r)$

JEE ADVANCED INSIGHTS

1. Single Atom vs Gas Sample Spectral Line Count:
For a gas sample containing millions of excited atoms in state n, all possible transition paths occur simultaneously, yielding $N = n(n - 1) / 2$ spectral lines. However, for a SINGLE isolated excited atom in state n, the maximum number of emitted spectral lines is only $(n - 1)$, corresponding to a single step-by-step cascade down the energy ladder (e.g., 4→3→2→1 gives 3 photons).

2. High Quantum Number Limit (Correspondence Principle):
For transitions between very high adjacent energy levels (large n, e.g., n = 200 → 199), the frequency of the emitted photon $\nu_{photon} = (E_n - E_{n-1}) / h$ approaches the classical orbital revolution frequency $f_n = v_n / (2\pi r_n) \propto Z^2 / n^3$. This illustrates Bohr's Correspondence Principle, bridging quantum jumps to classical radiation at macroscopic quantum numbers.

3. Recoil Momentum of Atom During Photon Emission:
When a free stationary atom emits a photon of energy $E_{photon} = h * \nu$, conservation of momentum requires the atom of mass M to recoil in the opposite direction with momentum $p_{recoil} = h / \lambda = E_{photon} / c$.
The recoil kinetic energy of the atom is $K_{recoil} = p_{recoil}^2/(2M) = E_{photon}^2/(2Mc^2)$.
Consequently, the actual emitted photon energy is slightly less than the orbital energy level gap ΔE: $\Delta E = E_{photon} + K_{recoil}$.

HIGH-YIELD REVISION SHEET

Key Formulas:
- Rydberg Formula: $1 / \lambda = R * Z^2 * [(1 / n_1^2) - (1 / n_2^2)]$
- Series Limit (Shortest Wavelength): $\lambda_{min} = n_1^2 / (R * Z^2)$
- Longest Wavelength: $1 / \lambda_{max} = R * Z^2 * [(1 / n_1^2) - (1 / (n_1 + 1)^2)]$
- Total Emitted Lines (Gas Sample): $N = (N_2 - N_1) * (N_2 - N_1 + 1) / 2$
- Total Emitted Lines (To Ground State): $N = n * (n - 1) / 2$
- Hypothetical Atom Method: $F = -dU/dr$; $m v^2 / r = F(r)$; $m v r = nh / (2\pi)$

Key Observations:
1. Lyman series lies in UV; Balmer lies in Visible; Paschen/Brackett/Pfund lie in IR.
2. H_α line is $n_2 = n_1 + 1$; H_β line is $n_2 = n_1 + 2$; H_γ line is $n_2 = n_1 + 3$.
3. Series limit corresponds to transition from $n_2 = \infty$ to n_1.
4. Bohr model fails for multi-electron atoms and cannot explain Zeeman or Stark effects.
5. For high quantum numbers n, photon emission frequency approaches classical orbital frequency $f_n \propto Z^2 / n^3$.

NUCLEAR PHYSICS

Nuclear Basics

Representation and Composition:

- An atomic nucleus is represented as X_Z^A, where X is the chemical symbol of the element, Z is the atomic number, and A is the mass number.
- Atomic number Z represents the total number of protons in the nucleus.
- Mass number A represents the total number of nucleons (protons + neutrons). Number of neutrons inside the nucleus is N = A – Z.

Nuclear Radius (R):

- The radius of a nucleus depends on its mass number A according to the empirical relation:

  $R = R_0 * A^{(1/3)}$.

  where R_0 is a universal nuclear constant equal to 1.2 fm = 1.2×10^{-15} m.

Nuclear Volume (V):

- Assuming a spherical shape, nuclear volume is $V = (4/3)\pi R^3 = (4/3)\pi(R_0^3 A)$.
- Nuclear volume is directly proportional to mass number A ($V \propto A$).

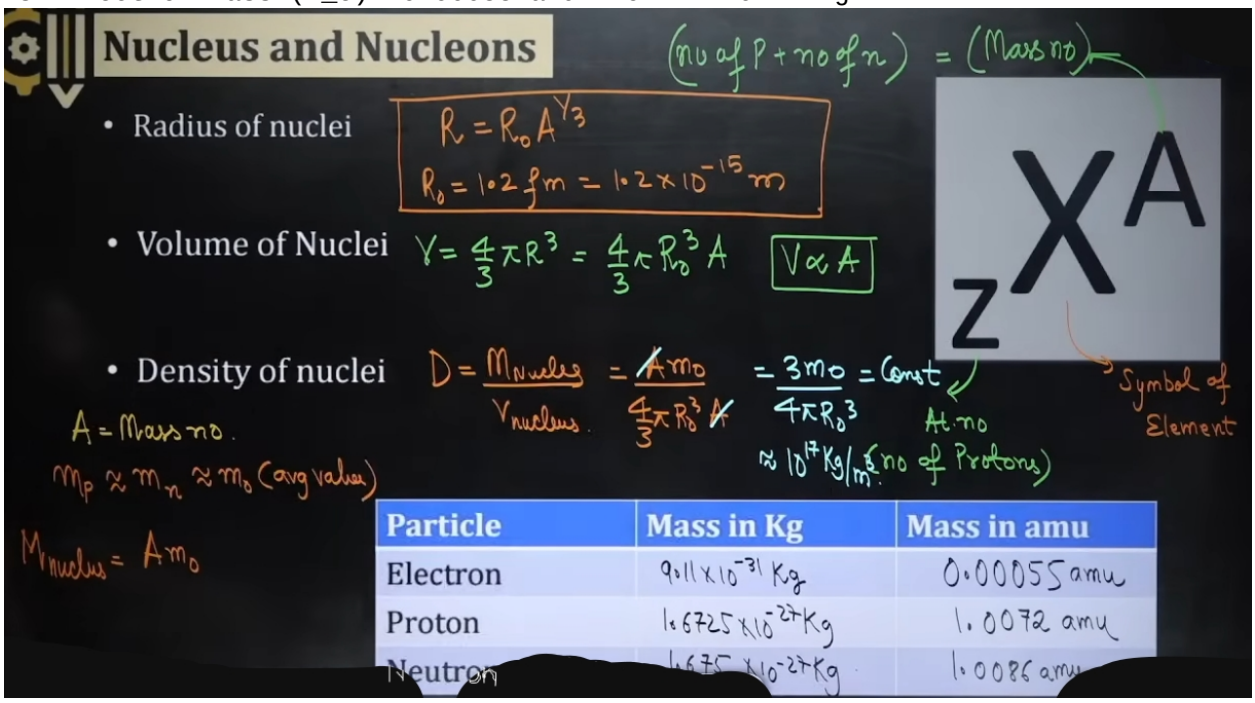
Universal Nuclear Density (ρ):

- Nuclear density is defined as nuclear mass divided by nuclear volume:

  $\rho = Nuclear\ Mass / Nuclear\ Volume = (A * m_nucleon) / [\ (4/3) * \pi * R_0^3 * A\ ] = (3 * m_nucleon) / (4 * \pi * R_0^3)$.
- Notice that mass number A cancels out completely from the density expression.
- Nuclear density is strictly CONSTANT for all nuclei regardless of mass number A or atomic number Z.
- Numerical Value: $\rho \approx 2.3 \times 10^{17}$ kg/m³, which is exceptionally high compared to atomic density.

Nucleon Rest Masses:

1. Proton Mass (m_p): 1.00727 amu = 1.6725×10^{-27} kg.
2. Neutron Mass (m_n): 1.00866 amu = 1.675×10^{-27} kg.
3. Electron Mass (m_e): 0.00055 amu = 9.1×10^{-31} kg.



Properties of Nuclear Forces

Strongest Fundamental Force:

- The nuclear force is the strongest force in nature, roughly 100 times stronger than the electromagnetic force and 10^{38} times stronger than the gravitational force.

Short-Range Nature:
- The nuclear force operates strictly over short sub-atomic distances within the nucleus (~ 1 to 2 fm) and drops rapidly to zero beyond nuclear dimensions.

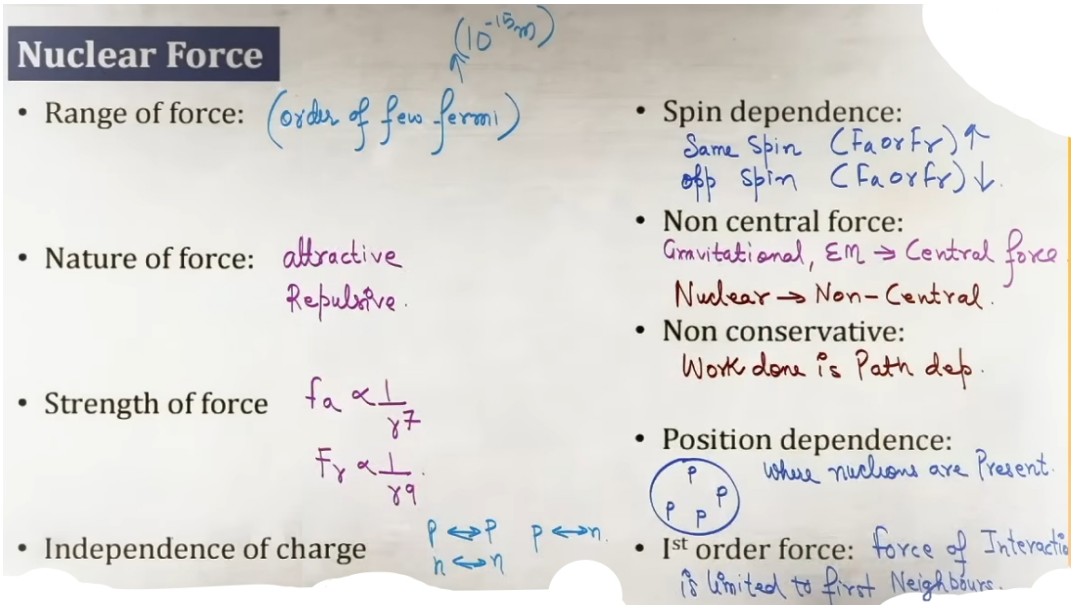
Distance Dependence (Attractive vs Repulsive):
- At distances r > 0.8 fm, the nuclear force is strongly attractive, binding nucleons together.
- At extremely short distances r < 0.8 fm, the force becomes strongly repulsive, preventing the collapse of nucleons into a single point.
- Mathematical Dependence: Attractive force component scales as $F_{attr} \propto 1 / r^7$, while repulsive component scales as $F_{rep} \propto 1 / r^9$.
- Non-Inverse-Square Law: The nuclear force does NOT follow the classical inverse-square law ($1/r^2$).

Charge Independence:
- The nuclear force acts equally between any two nucleons regardless of their electric charge.
- Force magnitudes satisfy: $F_{pp} = F_{nn} = F_{pn}$.

Spin Dependence:
- Nuclear forces depend on the relative spin orientation of interacting nucleons.
- Nucleons with parallel spins experience a stronger attractive force than nucleons with anti-parallel spins.

Non-Central and Non-Conservative Character:
- The nuclear force does not act strictly along the straight line joining nucleon centers (non-central).
- Work done by nuclear forces depends on the path taken (non-conservative).

Saturation Property / First-Neighbor Interaction:
- A nucleon interacts only with its immediate first-neighbor nucleons rather than all nucleons in the nucleus.
- Adding more nucleons to a large nucleus does not increase the binding force on a specific internal nucleon.

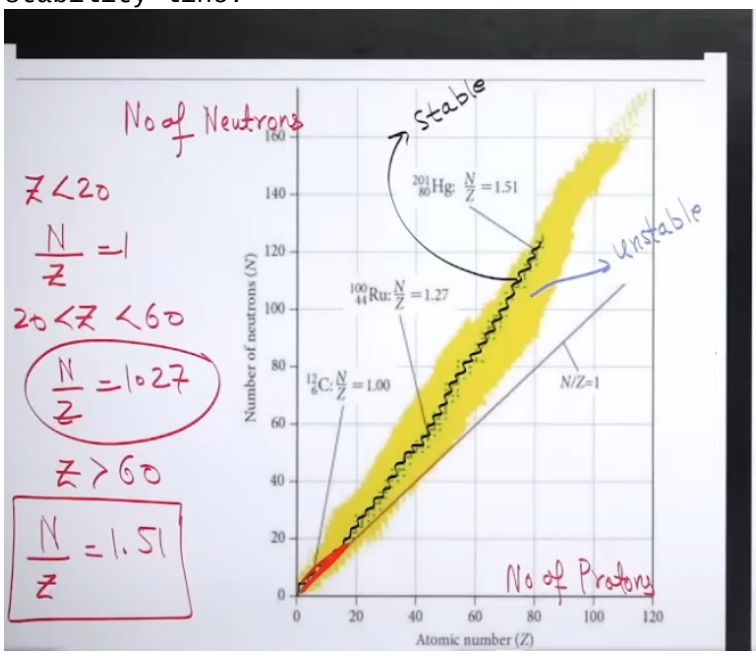


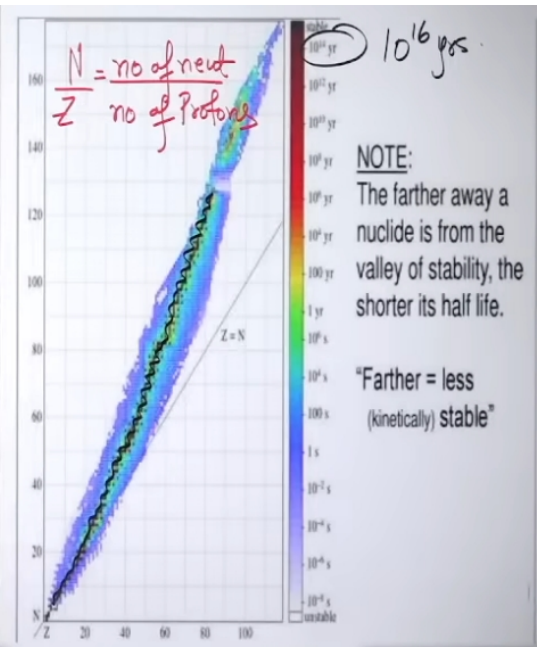
Nuclear Stability and N/Z Ratio
Stability Belt / N/Z Ratio Graph:
- Plotting the number of neutrons N against the number of protons Z reveals a narrow "valley of stability".
  1. Light Nuclei (Z < 20): Stable when N/Z ≈ 1.0 (equal protons and neutrons, e.g., C^{12}, O^{16}).
  2. Medium Nuclei (20 < Z < 60): Stable when N/Z ≈ 1.27.
  3. Heavy Nuclei (Z > 60): Stable when N/Z ≈ 1.51.

Reason for Increased N/Z in Heavy Nuclei: As Z increases, electrostatic repulsion between protons scales as Z^2, requiring excess neutrons to supply attractive nuclear force without adding disruptive electrostatic repulsion.

Instability and Decay Trends:

Nuclei lying above or below the stability belt are unstable and spontaneously undergo radioactive decay (α, β^-, β^+) to adjust their N/Z ratio toward the stability line.





Mass Defect(Δm), Mass-Energy Equivalence($E=mc^2$), Binding Energy, and BE/A Curve

Mass-Energy Equivalence:

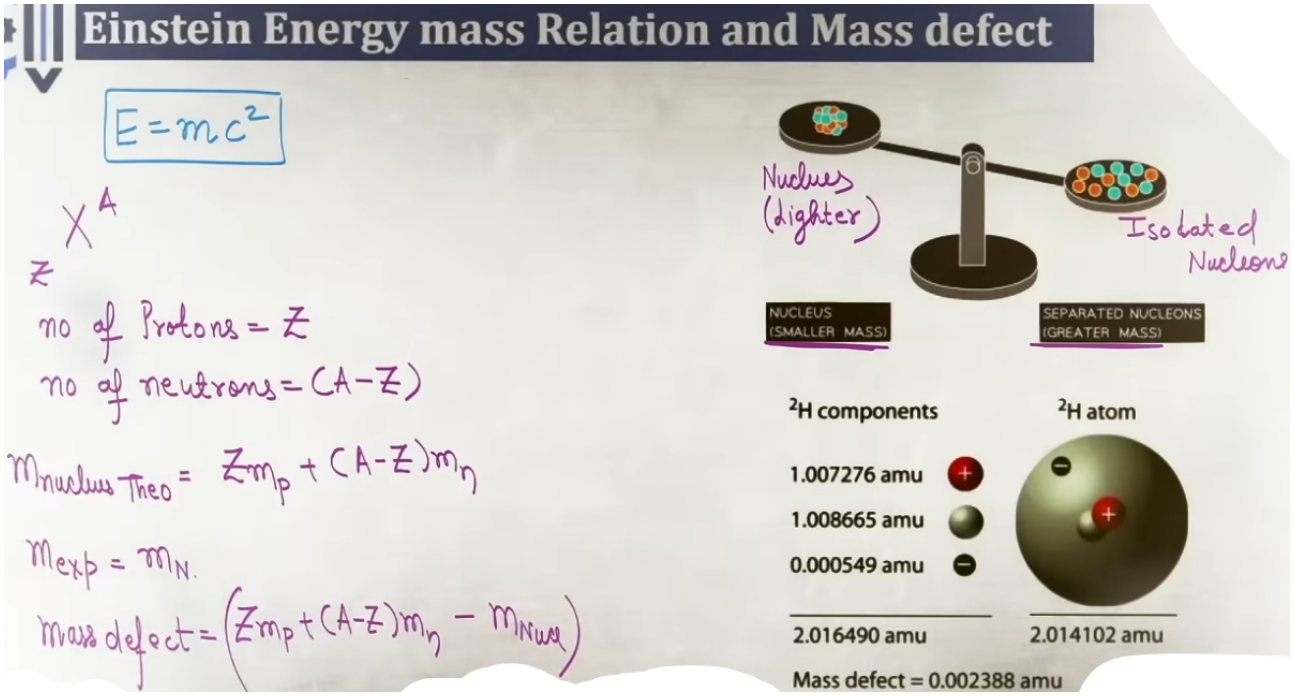
- Einstein's principle states that mass and energy are interconvertible: **$E = mc^2$**

Mass Defect (Δm):

- The rest mass of a stable, bound nucleus is always LESS than the sum of the rest masses of its constituent free protons and neutrons.
- Mass Defect Formula:

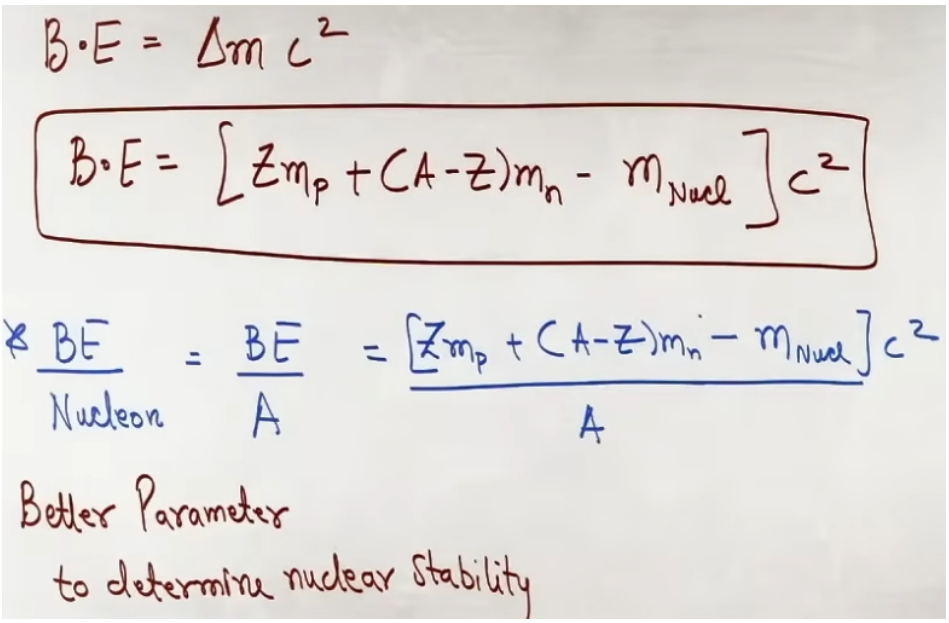
$\Delta m = [\ Z*m_p + (A-Z)*m_n\ ] - M_{nucleus}$

where m_p is proton mass, m_n is neutron mass, and $M_{nucleus}$ is the measured nuclear mass.



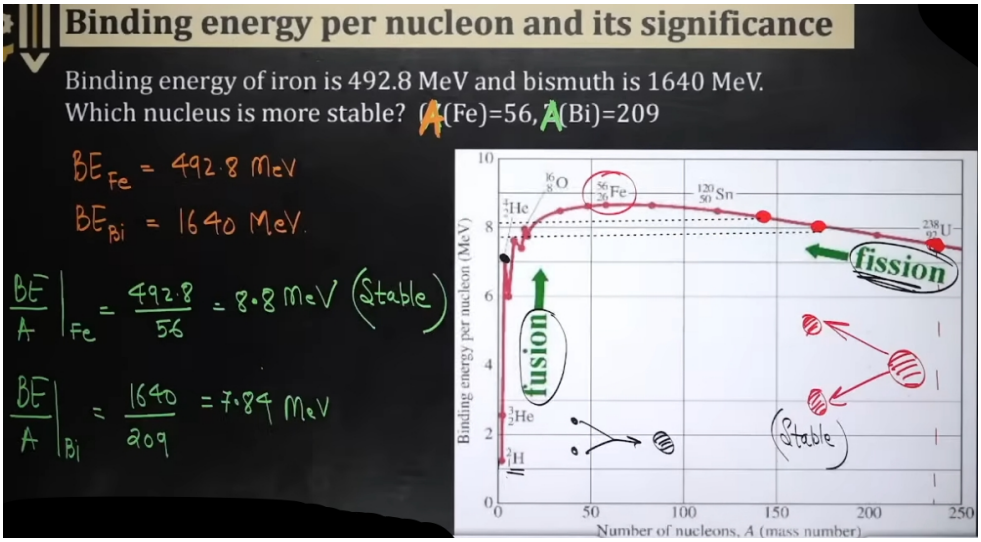
Binding Energy (BE):
- The energy released when free nucleons come together from infinity to form a bound nucleus, or the energy required to completely disassemble a nucleus into its constituent free nucleons.
  Formula: **BE = Δm * c²**.
  Energy Equivalence Shortcut: **1 amu (u) = 931.5 MeV**.
  BE (in MeV) = Δm (in amu) * 931.5 MeV.

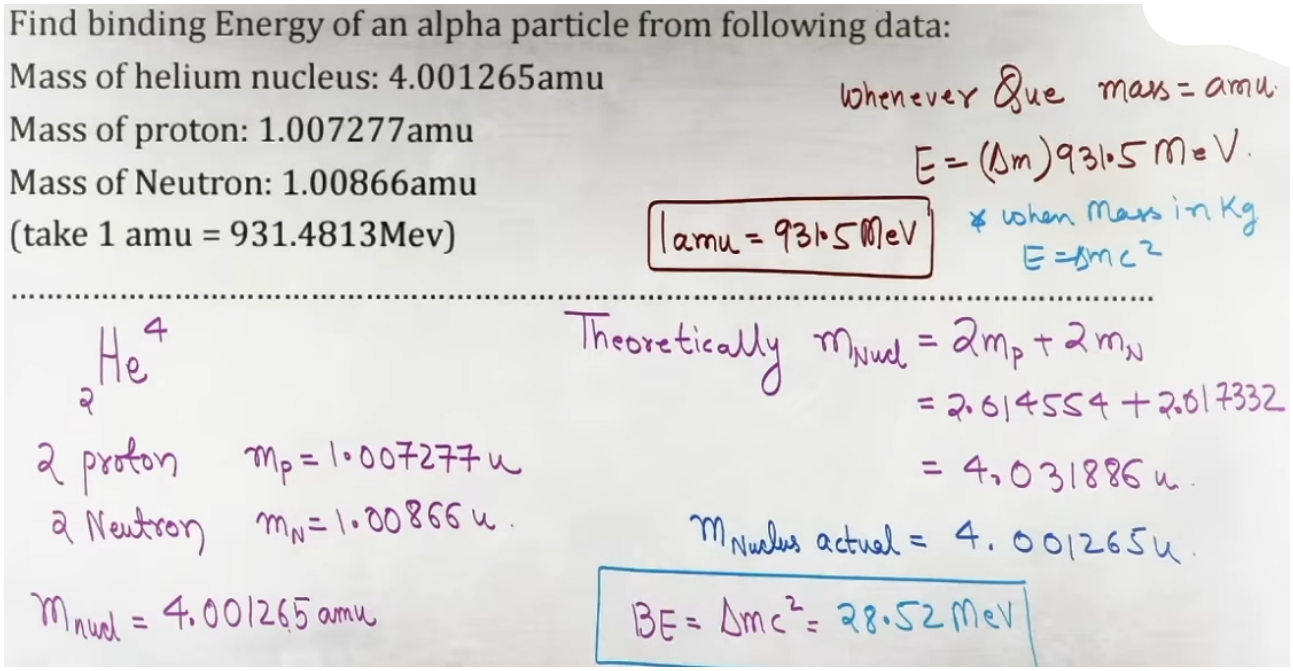
Binding Energy per Nucleon (BE / A):
  BE / A = Total Binding Energy / Mass Number A.
  This is the true measure of nuclear stability (higher BE/A means greater stability).



Binding Energy per Nucleon Curve Analysis:
  1. Peak Stability: Fe^{56} has the highest BE/A ≈ 8.8 MeV/nucleon, making it the most stable nucleus in nature.
  2. Heavy Nuclei Region (A > 200): BE/A drops gradually to ~ 7.6 MeV/nucleon due to cumulative electrostatic repulsion between protons.
  3. Light Nuclei Region (A < 20): BE/A is low, but shows sharp local peaks for even-even nuclei like He^{4} (7.07 MeV/nucleon), C^{12}, and O^{16}.

Nuclear Fission and Fusion Mechanisms:
- Fission: A heavy unstable nucleus (A > 200) splits into two medium-sized fragments with higher BE/A, releasing energy.
- Fusion: Two light nuclei (A < 20) fuse together to form a heavier, more tightly bound nucleus with higher BE/A, releasing massive energy.





Nuclear Fission, Fusion, and Q-Value

Conservation Laws in Nuclear Reactions:

- In any nuclear reaction, total charge number Z, total mass number A, linear momentum, angular momentum, and total energy are strictly conserved.

Q-Value Definition:

The net energy released or absorbed in a nuclear reaction.

Formula using Masses:

Q = [Mass of Reactants - Mass of Products] * c².

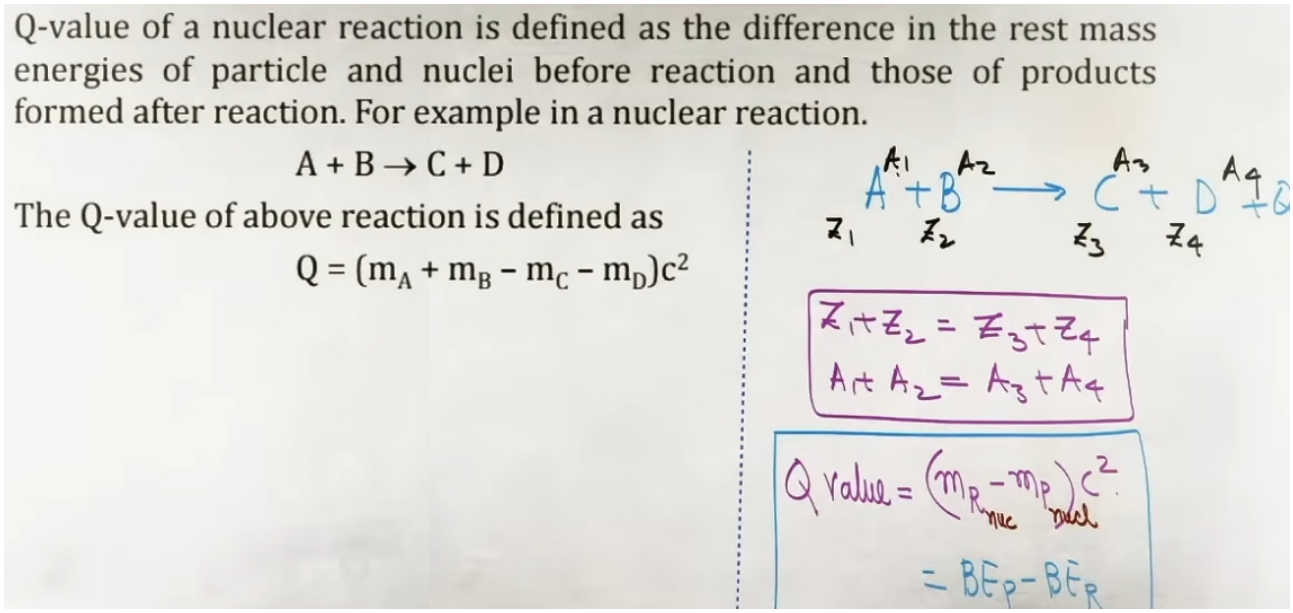
Q = [(M_A + M_B) - (M_C + M_D)] * c².

Formula using Binding Energies:

Q = Total BE of Products - Total BE of Reactants.

Reaction Types based on Q-Value:

- Exothermic Reaction (Q > 0): Mass of reactants > Mass of products. Energy is released.
- Endothermic Reaction (Q < 0): Mass of reactants < Mass of products. Energy must be supplied as kinetic energy to drive the reaction.



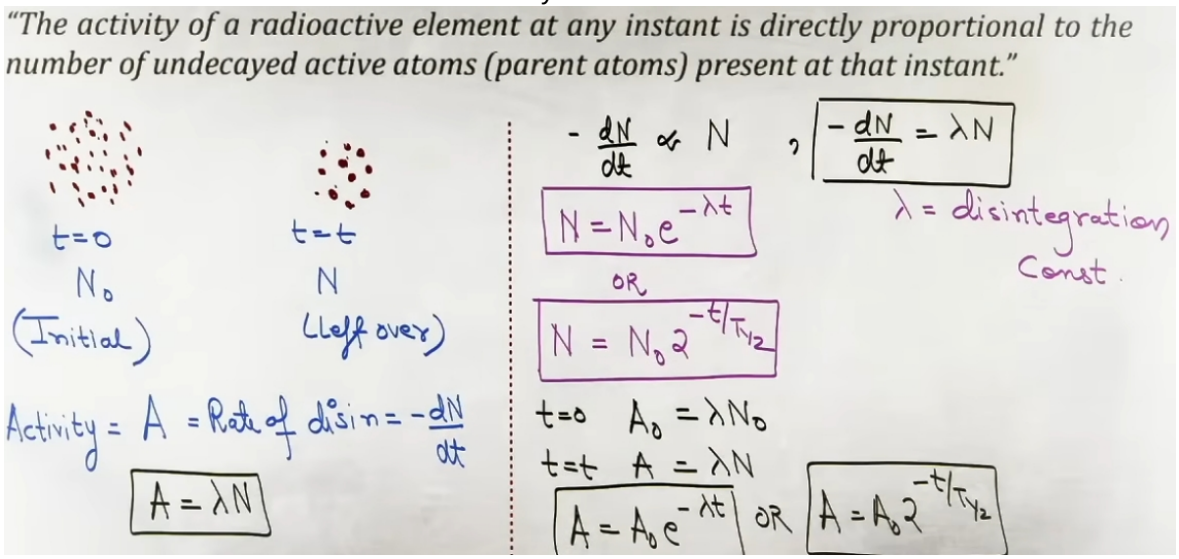
Radioactivity and Decay Laws

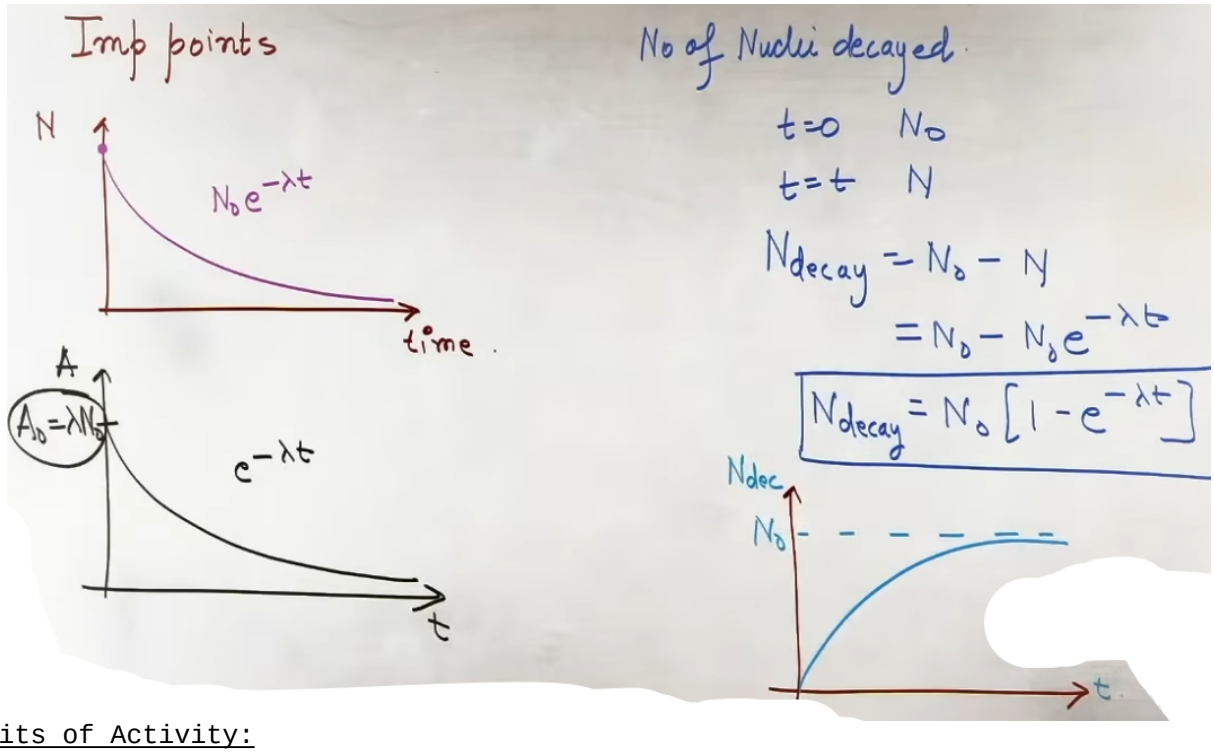
Definition:

- Radioactivity is the spontaneous, random nuclear transformation of an unstable nucleus into another nucleus with the emission of particles or radiation.
- It is purely a nuclear phenomenon, completely independent of external physical or chemical conditions like temperature, pressure, or chemical bonding.

Rate of Decay / Activity (A):

- Activity A is the number of nuclear disintegrations occurring per unit time:
  A = -dN / dt = λ * N.
  where λ is the decay constant (disintegration constant), and N is the instantaneous number of active undecayed nuclei.





Units of Activity:

1 Becquerel (Bq) = 1 decay per second (SI Unit).
1 Curie (Ci) = 3.7 × 10^{10} Bq.
1 Rutherford (Rd) = 10^{6} Bq.

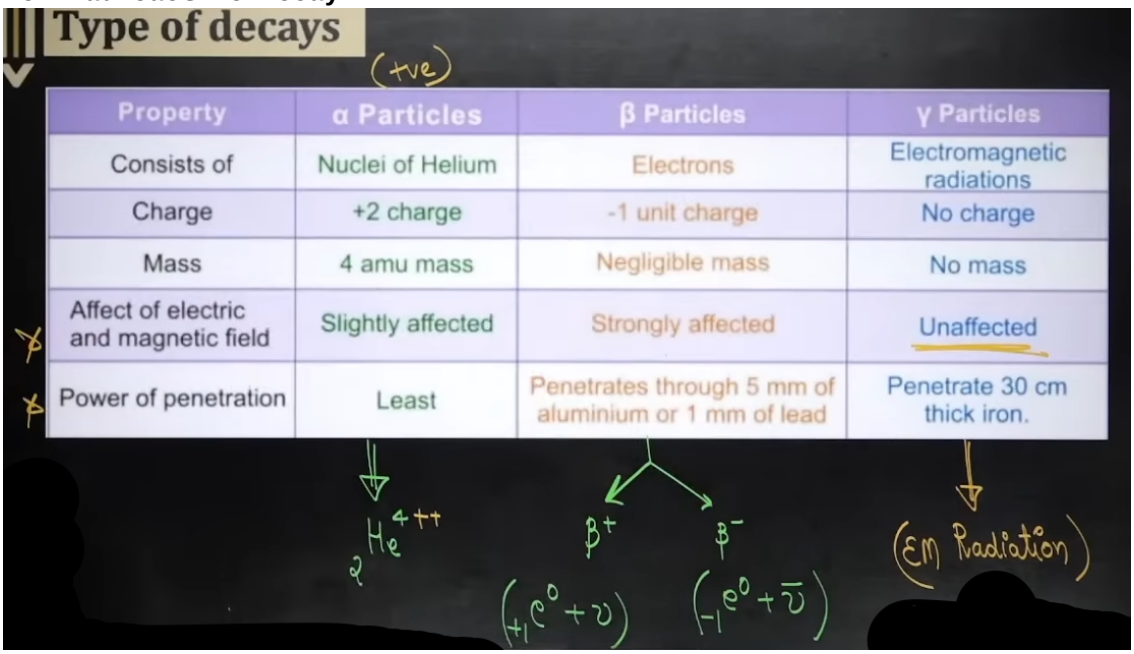
Integrated Decay Laws:

1. Undecayed Active Nuclei at time t: $N(t) = N_0 * e^{(-\lambda * t)}$.
2. Activity at time t: $A(t) = A_0 * e^{(-\lambda * t)}$ where $A_0 = \lambda * N_0$.
3. Decayed Nuclei at time t: $N_{decayed}(t) = N_0 - N(t) = N_0 * (1 - e^{(-\lambda * t)})$.

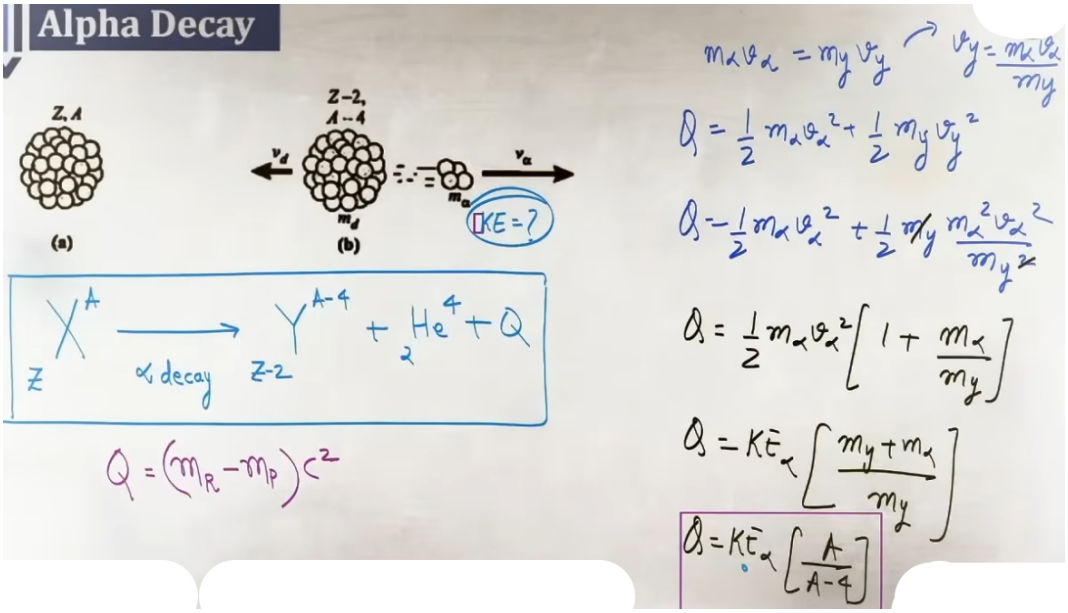
Half-Life (T_1/2):
  The time required for half of the initial active nuclei to decay.
  Formula: **T_1/2 = ln(2) / λ = 0.693 / λ.**
  Fraction remaining after n half-lives (where t = n * T_1/2):
      N(t) = N₀ / 2ⁿ = N₀ / 2^(t / T_1/2).
  Activity remaining after n half-lives:
      A(t) = A₀ / 2ⁿ = A₀ / 2^(t / T_1/2).

Mean Life / Average Life (τ):
  The average lifespan of all nuclei in a sample.
  Formula: τ = 1 / λ = T_1/2 / ln(2) ≈ 1.44 * T_1/2.

Modes of Radioactive Decay

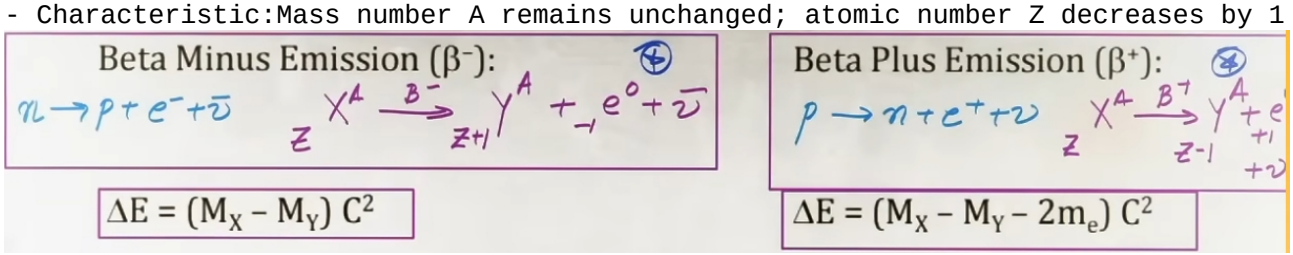


1. Alpha (α) Decay:
  Emission of a Helium nucleus He_2^4 (alpha particle) from a heavy nucleus.
  General Reaction: **X_Z^A → Y_(Z-2)^(A-4) + He_2^4 + Q.**
  Kinetic Energy Distribution:
  By momentum conservation (p_α = p_Y) in a stationary parent decay:
  Recoil kinetic energy of Alpha Particle: K_α = Q * (A - 4) / A.
  Recoil kinetic energy of Daughter Nucleus: K_Y = Q * 4 / A.

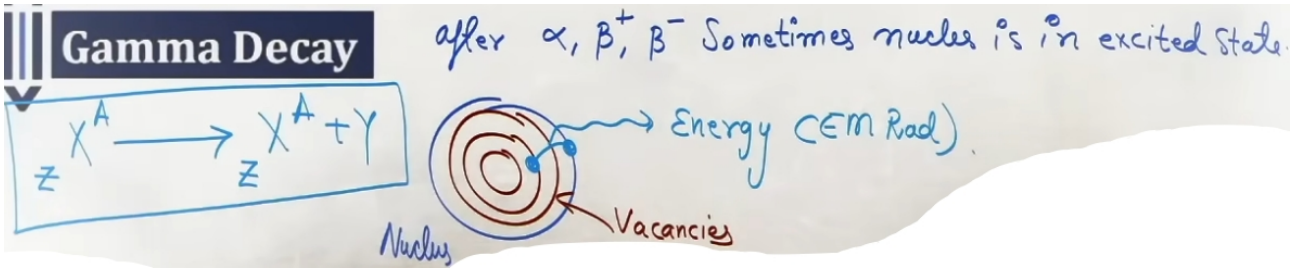


2. Beta-Minus (β⁻) Decay:
- Conversion of a neutron into a proton inside a neutron-rich nucleus: n → p + e⁻ + ν_bar.
- General Reaction: **X_Z^A → Y_(Z+1)^A + e_(-1)^0 + ν_bar (anti-neutrino)**.
- Characteristic:Mass number A remains unchanged; atomic number Z increases by 1
- Continuous β-energy spectrum is explained by sharing of Q-value between electron and anti-neutrino.

3. Beta-Plus (β⁺) Decay:
- Conversion of a proton into a neutron inside a proton-rich nucleus: p→n+ e⁺+ ν
- General Reaction: **X_Z^A → Y_(Z-1)^A + e_1^0 + ν (neutrino)**.
- Characteristic:Mass number A remains unchanged; atomic number Z decreases by 1



4. Gamma (γ) Decay:
- De-excitation of an excited nuclear energy level to a lower energy state by emitting high-energy photons.
- General Reaction: **X_Z^(A*) → X_Z^A + γ**.
- Characteristics: Neither atomic number Z nor mass number A changes.



Property Comparison of Radiations:
Ionization Power: α > β > γ (Alpha particles produce dense ionization).
Penetration Power: γ > β > α (Gamma rays penetrate up to 30 cm of iron).
Electric / Magnetic Field Deflection: Alpha and Beta particles are deflected in opposite directions; Gamma rays pass completely undeflected.

Q. In a radioactive decay chain, the initial nucleus is ${}^{232}_{90}Th$. At the end, there are 6α-particles and 4β-particles which are emitted. If the end nucleus is X, A and Z are given by **(Main 2019, 12 Jan II)**

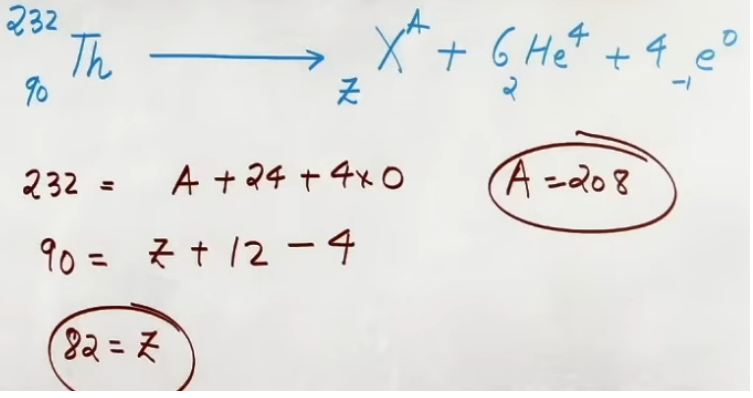
Ans : B

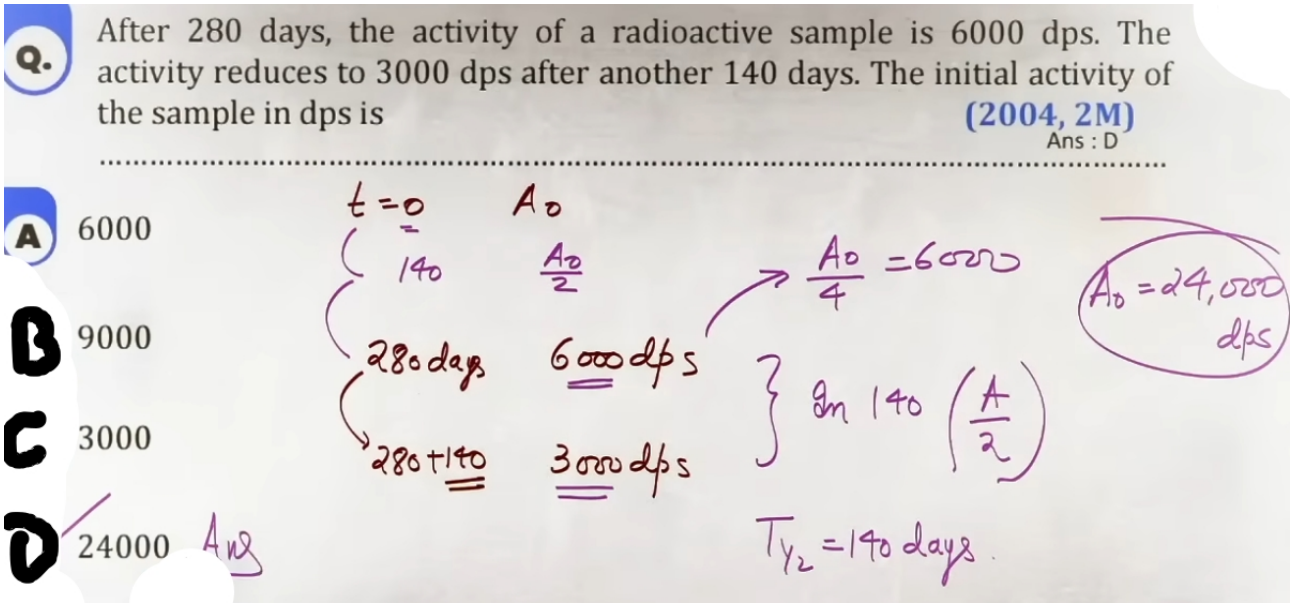
A. A = 202; Z = 80

B. A = 208; Z = 82 Ans

C. A = 200; Z = 81

D. A = 208; Z = 80





INTUITION AND CONCEPTUAL UNDERSTANDING

- Mass Defect Analogy (The Energy Sacrifice for Stability):
  Free nucleons at rest have high potential energy. When brought together into a compact nucleus, they fall into a deep attractive potential energy well. To remain bound, the system must shed energy equal to BE = Δm * c^2. The nucleus sacrifices a tiny fraction of its total mass as emitted energy. Reversing the process requires returning that exact energy to free the nucleons.

- Saturation of Nuclear Force Intuition:
  Unlike gravity or electrostatic force where every particle interacts with all other particles, nuclear force saturates. A nucleon inside a large nucleus is glued only to its immediate neighbors. Adding more nucleons further away does not increase the binding force on that nucleon, explaining why BE/A is roughly constant (~ 8 MeV) over a broad range of mass numbers A = 30 to 170.

- Radioactivity Randomness (Popcorn Analogy):
  In a bag of popcorn over heat, you cannot predict which specific kernel will pop next, but you know the overall popping rate. Similarly, radioactive decay is entirely probabilistic. For a given nucleus, decay is a matter of pure chance, but for a sample of 10^{20} nuclei, rate laws N(t) = N_0 e^(-λt) hold with extreme precision.

- Recoil Partition in Alpha Decay:
  When a stationary heavy nucleus fires an alpha particle, momentum conservation requires $p_\alpha = p_Y$. Since kinetic energy is K = p^2 / (2m), kinetic energy splits inversely with mass. Because the daughter nucleus is much heavier than the alpha particle (mass ratio ~ (A-4)/4), the light alpha particle carries away the vast majority (~ 98%) of the reaction energy Q.

FORMULA SHEET

- Nuclear Radius & Density:
  R = 1.2 × 10^{-15} * A^(1/3) m
  ρ ≈ 2.3 × 10^{17} kg/m^3 (Universal Constant)

- Mass Defect & Binding Energy:
  Δm = [Z * m_p + (A - Z) * m_n] - $M_{nucleus}$
  BE (in MeV) = Δm (in amu) * 931.5 MeV
  BE / A = BE / A

- Q-Value of Nuclear Reaction:
  Q = [Mass(Reactants) - Mass(Products)] * 931.5 MeV
  Q = Total BE(Products) - Total BE(Reactants)

- Decay Law & Activity Formulas:
  Activity: $A = -dN / dt = \lambda * N$
  $N(t) = N_0 * e^{(-\lambda * t)}$
  $A(t) = A_0 * e^{(-\lambda * t)}$
  $N_{decayed}(t) = N_0 * (1 - e^{(-\lambda * t)})$

- Half-Life & Mean Life:
  $T_{1/2} = 0.693 / \lambda$
  Remaining fraction after n half-lives: $N = N_0 / 2^n$ where $n = t / T_{1/2}$
  Mean Life: $\tau = 1 / \lambda = 1.44 * T_{1/2}$

- Alpha Decay Kinetics:
  $K_\alpha = Q * (A - 4) / A$
  $K_{daughter} = Q * 4 / A$

Comparison Table of Radioactive Decay Modes:

Property	Alpha (α) Decay	Beta-Minus (β^-) Decay	Beta-Plus (β^+) Decay	Gamma (γ) Decay
Identity	Helium nucleus (He_2^4)	Fast electron (e^-)	Positron (e^+)	Photons (EM radiation)
Nuclear Reaction	$X_Z^A \rightarrow Y_{(Z-2)}^{(A-4)} + \alpha$	$X_Z^A \rightarrow Y_{(Z+1)}^A + e^- + \nu_{bar}$	$X_Z^A \rightarrow Y_{(Z-1)}^A + e^+ + \nu$	$X_Z^{(A*)} \rightarrow X_Z^A + \gamma$
Atomic Number Z	Decreases by 2	Increases by 1	Decreases by 1	Unchanged
Mass Number A	Decreases by 4	Unchanged	Unchanged	Unchanged
Ionization Power	Highest	Moderate	Moderate	Lowest
Penetration Power	Lowest	Moderate	Moderate	Highest
Field Deflection	Deflected	Deflected opposite to α	Deflected with α	No Deflection

EXAMPLES AND APPLICATIONS

EXAMPLE 1: Binding Energy of Alpha Particle
Problem: Calculate the binding energy of an alpha particle (He_2^4) given: m_p = 1.00727 amu, m_n = 1.00866 amu, and M_{He} = 4.00150 amu.
Solution:
1. Constituent Nucleons: 2 protons and 2 neutrons.
2. Mass of free nucleons:
   $m_{free} = 2 * (1.00727) + 2 * (1.00866) = 2.01454 + 2.01732 = 4.03186$ amu.
3. Mass Defect:
   $\Delta m = 4.03186 - 4.00150 = 0.03036$ amu.
4. Binding Energy:
   $BE = 0.03036 * 931.5$ MeV ≈ 28.28 MeV.
5. BE per Nucleon:
   $BE / A = 28.28 / 4 \approx 7.07$ MeV/nucleon.

EXAMPLE 2: Radioactive Decay Activity Reduction
Problem: A radioactive sample has an initial activity of 1600 counts/sec. After 8 seconds, its activity drops to 100 counts/sec. Find the half-life of the sample.
Solution:
1. Activity ratio: $A(t) / A_0 = 100 / 1600 = 1 / 16$.
2. Express as power of 2: $1 / 16 = (1 / 2)^4$.
3. Number of half-lives elapsed n = 4.
4. Total time $t = n * T_{1/2} \Rightarrow 8$ sec $= 4 * T_{1/2}$.
5. Half-life $T_{1/2} = 8 / 4 = 2$ seconds.

JEE ADVANCED INSIGHTS

1. Universal Constant Density of All Nuclei:
Because nuclear radius $R = R_0 * A^{(1/3)}$, nuclear volume scales strictly as $V \propto A$. Consequently, nuclear mass divided by volume yields a constant density $\rho \approx 2.3 \times 10^{17}$ kg/m³ for ALL elements from Hydrogen to Uranium.

2. Recoil Kinetic Energy in Alpha Decay:
Alpha decay energy Q is shared between the alpha particle and daughter nucleus inversely proportional to their masses: $K_\alpha = Q(A-4)/A$ and $K_{daughter} = Q(4)/A$. Never assume the alpha particle takes 100% of the Q-value!

3. Continuous Beta Spectrum and Anti-Neutrino:
In beta decay, energy Q is shared among three products (daughter nucleus, beta particle, and neutrino/anti-neutrino). Because energy is shared three ways, emitted beta particles possess a continuous spectrum of kinetic energies from 0 to K_{max}.

HIGH-YIELD REVISION SHEET

Key Formulas:
- Nuclear Radius: $R = 1.2 \times 10^{-15} * A^{(1/3)}$ m
- Nuclear Density: $\rho \approx 2.3 \times 10^{17}$ kg/m³ (Constant)
- Mass Defect: $\Delta m = [Z * m_p + (A - Z) * m_n] - M_{nucleus}$
- Binding Energy: $BE = \Delta m * 931.5$ MeV
- Q-Value: $Q = [M_{reactants} - M_{products}] * 931.5$ MeV = $BE_{products} - BE_{reactants}$
- Decay Rate / Activity: $A = -dN / dt = \lambda * N$
- Decay Equation: $N(t) = N_0 e^{(-\lambda t)} = N_0 / 2^n$ where $n = t / T_{1/2}$
- Half-Life & Mean Life: $T_{1/2} = 0.693 / \lambda$; $\tau = 1 / \lambda = 1.44 * T_{1/2}$
- Alpha Recoil Energy: $K_\alpha = Q * (A - 4) / A$

Key Observations:
1. Universal nuclear density is independent of A and Z (~ 2.3×10^{17} kg/m³).
2. Fe^{56} has highest BE/A ≈ 8.8 MeV/nucleon (most stable).
3. Heavy nuclei undergo fission; light nuclei undergo fusion to reach peak BE/A.
4. Radioactivity is a spontaneous, random, nuclear process independent of external physical/chemical conditions.
5. Ionization power: α > β > γ; Penetration power: γ > β > α.

LAST-MINUTE REVISION TABLE:

Parameter / Process	Governing Equation	Key Fact / Trap
Nuclear Radius	$R = R_0 A^{(1/3)}$	Volume $V \propto A$
Nuclear Density	$\rho \approx 2.3 \times 10^{17}$ kg/m³	Strictly constant for all nuclei
Binding Energy	$BE = \Delta m * 931.5$ MeV	Measure of nuclear stability
Peak Stability	Fe^{56} (BE/A ≈ 8.8 MeV)	Highest BE/A in nature
Radioactive Decay	$N(t) = N_0 / 2^n$	$n = t / T_{1/2}$ (no. of half-lives)
Half-Life & Mean Life	$T_{1/2} = 0.693 / \lambda$; $\tau = 1.44 T_{1/2}$	$\tau = 1 / \lambda$
Alpha Decay Kinetics	$K_\alpha = Q * (A - 4) / A$	Alpha carries ~ 98% of Q-value
Penetration Power	γ > β > α	Gamma rays penetrate up to 30 cm iron